Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

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Abstract

Mutual information (MI) measures how much observing one variable reduces uncertainty about another. This thesis considers whether MI between two categorical variables differs across independent populations. A Normal Wald test provides one approach, but small or sparse samples can distort the estimated difference and its standard error, producing unreliable false-positive rates and power.

This thesis develops an Expanded Welch–Satterthwaite test that retains Wald's estimated difference and standard error and uses a Student t distribution. Its degrees of freedom depend on how much the estimated MI variance changes between samples. The thesis derives a first-order approximation to that variability by accounting for changes in cell probabilities, their margins and the resulting MI values.

Simulations first check the moment approximations used in the derivation. They then compare false-positive rates and power with Normal Wald, a sample-size degrees-of-freedom rule and an independently fitted moment reference. Power is also compared after matching null rejection rates. An online-shopping dataset illustrates interpretation on observed data.

The mean and sampling variance approximations can have substantially different errors. Expanded Welch improves calibration in selected small-sample populations and can improve power at matched false-positive rates. However, changing the cutoff does not correct sparse MI bias or inaccurate uncertainty estimates. The results identify where MI-specific degrees of freedom are useful and the estimation problems that limit their application.

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Chapter 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature X may carry different amounts of information about a class label Y in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under H_0, the populations have equal MI. However, this does not mean that their joint probability tables are the same, as two populations can have equal MI while having different probability tables.

In practice, MI is estimated from observed data. Two samples can produce different MI estimates even when the population MI values are equal, because their observed counts vary from sample to sample. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be

inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student t distribution instead of the standard normal distribution. Choosing its degrees of freedom requires estimating how much the MI variance estimates change from sample to sample. For MI, this calculation must account for the fact that changing a cell probability also changes its row and column totals and the MI values calculated from them. This thesis derives how these probability changes affect the estimated MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student t distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. This can reduce false positives, but it can also reduce statistical power: the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation measures the resulting false-positive rates and nominal power, then compares power after matching null rejection rates through independent simulation. This asks whether the sample-dependent degrees of freedom change discrimination as well as the overall rejection rate. The frequency of usable p-values is reported alongside these rates.

1.2 Research questions

The thesis addresses two research questions:

1. How accurately do the derived approximations predict the mean and sampling variance of the estimated MI variance, and what degrees of freedom do they imply near and away from independence?
2. How do Expanded Welch's false-positive rates and power compare with Normal Wald and a sample-size degrees-of-freedom comparator?

The analytical derivation gives the variance sensitivity and component degrees of freedom. Controlled simulations check its moment approximations separately, before comparing the completed rejection rules. Independent moment fitting provides a fixed simulation reference. Power is measured both at nominal cutoffs and after

independent null calibration, so the comparison distinguishes a change in false positives from a change in power at matched null rejection.

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance, including the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student t distribution.
3. **Empirical contribution.** Controlled simulations examine how sample size, table size, category frequencies and population MI affect false-positive rates, power and the frequency of usable results. Supporting diagnostics assess errors in the MI difference and its standard error, and test the predicted mean and sampling variance of the variance estimator and the resulting component degrees of freedom. Additional supplementary checks look at further population constructions and sample allocation, while a timing comparison measures computational cost.
4. **Practical contribution.** The evaluation identifies when Expanded Welch brings false-positive rates closer to the intended 5% level than Normal Wald, and when estimation errors make both tests unreliable.

1.4 Scope and organisation

The scope of this thesis is the comparison of MI between two populations using categorical data. Each sample is summarised in a count table, and both tables use the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. The simulations compare Normal Wald, Expanded Welch, a sample-size degrees-of-freedom rule and an independently fitted moment reference, using two-sided tests at the 5% significance level.

The organisation of the thesis follows the development, evaluation and interpretation of the two-sample MI test. Chapter 2 reviews the background and literature.

Chapters 3 and 4 define the shared statistic and derive Expanded Welch. Chapter 5 describes the experiment, and Chapter 6 discusses the results. Chapter 6 also illustrates both tests through a real-world example using an online-shopping dataset. Chapters 7 and 8 discuss the implications, limitations and conclusions. The appendices provide supporting calculations, experimental settings and additional results.

Chapter 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts, whereas a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. The null hypothesis requires equal MI, not identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to n. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. In particular, rare cells can change both even when the categories and sample size are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the large-sample expansion of the plug-in estimator's bias has a positive leading term:

$$\text{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing MI estimates from two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably. Rare category combinations can then be poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. Chao and Shen (2003) instead adjust entropy for categories missing from the sample. Their method uses sample coverage, the total population probability of the categories that have been observed, and accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. Nemenman et al. (2002) combine priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data. This is the Nemenman–Shafee–Bialek (NSB) estimator. More recently, Piga et al. (2024) estimate the strength of a Dirichlet prior from the observed counts. This strength controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

The sampling pattern can also suggest a more targeted estimator. Hernández and Samengo (2019) consider a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. They use the well-estimated marginal

and model the conditional probabilities to estimate MI, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. Estimation errors need not cancel when MI estimates from two distinct populations are subtracted. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Comparing two MI estimates requires knowing how much each varies across repeated samples. We derive the large-sample approximation

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004).

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x-a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j} (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i} (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i} \left[(\log p_{i+} + 1) \left\{ \sum_{j} (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned}(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\ = \ell_P(i,j) - 1. \end{aligned} \tag{2.19}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \tag{2.20}$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j}(\widehat{p}_{ij} - p_{ij}) = \sum_{i,j}\widehat{p}_{ij} - \sum_{i,j}p_{ij} = 1 - 1 = 0. \tag{2.21}$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.22}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n}\sum_{a=1}^{n}\left\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\right\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\ell_P(i,j)\left[\frac{1}{n}\sum_{a=1}^{n}\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}\right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n}\sum_{a=1}^{n}\left[\sum_{i,j}\ell_P(i,j)\mathbf{1}(Z_a=(i,j)) - \sum_{i,j}p_{ij}\ell_P(i,j)\right] \\ &= \frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\,. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned}\mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j}p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j}p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned}\mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to testing MI differences

This section reviews methods for estimating uncertainty in MI and testing differences between two MI values. The question in this thesis is whether two populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

The Normal Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate, and independence of the two samples makes their covariance zero. Replacing the population variances by

estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

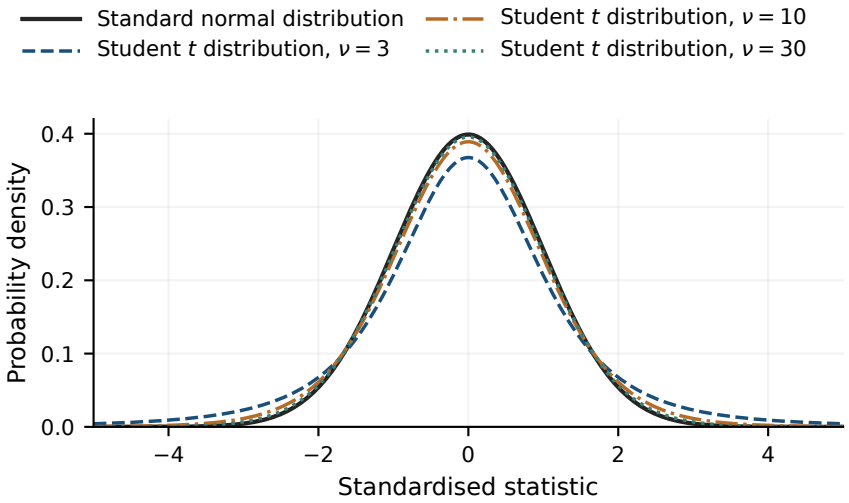


Figure 2.1: Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

This distinction matters because the size of a variance estimate and its stability answer different questions. The estimate $\widehat{V}(P)/n_P$ describes the expected spread of an MI estimate across samples. The variability of $\widehat{V}(P)$ describes how much that assessment of spread itself changes across samples. Two variance estimators with the same mean can have different variability. Satterthwaite's calculation uses both moments: greater variability relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example,

Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples, and Expanded Welch uses the variability of the estimated MI variance to choose degrees of freedom in the same way. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings.

Previous work on MI uncertainty and comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations P and Q.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student t distribution, using

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the

same observations. Their estimation errors are connected, because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. Expanded Welch also tracks how the estimated MI variance changes: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

An ordinary empirical bootstrap cannot sample an unobserved cell, because it assigns that cell zero probability. It can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test instead combines the two samples and shuffles their group labels. This gives an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Equal MI alone does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student t distribution rather than a standard normal distribution.

The experiment measures errors in the shared MI difference and standard error alongside false-positive rates and power. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection.

Chapter 3

The Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Miller, 1955; Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes,

their difference removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Under H_0, we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function.

Why use a Student t distribution?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the variability of that variance estimate to choose a Student t distribution instead. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples. The name Expanded Welch refers to this expansion of the MI variance as the table probabilities change.

4.1 Choosing degrees of freedom for Expanded Welch

The Welch–Satterthwaite equation combines the two estimated MI variance contributions (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The terms $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$ are the estimated variance contributions used to calculate the standard error for both Normal Wald and Expanded Welch in Chapter 3. The new variables being introduced are the component degrees of freedom ν_P and ν_Q. They describe how much each estimated MI variance varies between samples relative to its mean.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$

changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one MI variance estimate.

Satterthwaite approximates the nonnegative variance estimate $\widehat{V}(P)$ by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}. \tag{4.2}$$

The scale and degrees of freedom are chosen to match the mean and variance of $\widehat{V}(P)$.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_P}) = \nu_P, \qquad \mathrm{Var}(\chi^2_{\nu_P}) = 2\nu_P. \tag{4.3}$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.4}$$

Its variance is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P^2}\,\mathrm{Var}(\chi^2_{\nu_P}) \tag{4.5}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \tag{4.6}$$

Matching this quantity to $\mathrm{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.7}$$

The numerator depends on the mean of the estimated MI variance; the denominator measures how much that estimate fluctuates between samples. To first order, $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$.

4.2 How the MI variance changes with the table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.8}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.9}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.10}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.12}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.13}$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.14}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x,j=y\}-p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}-p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}-p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.15}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.16}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x,j=y)-p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 . \end{aligned} \tag{4.17}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.18}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.19}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.20}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.21}$$

Differentiating Equation (4.20) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.23}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 &= \sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &\quad + 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.24}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.25}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.26}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.27}$$

Substitution of Equation (4.15) gives

$$\begin{aligned}&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\ &\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\ &\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).\end{aligned} \tag{4.28}$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in

Equation (4.25) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.28) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\Big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\Big].
\end{aligned}} \tag{4.29}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.30}$$

The sensitivities average to zero under P, as shown in Appendix A. Their variation, not their average, determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{\delta_{Z_a^{(P)}} - P\right\}. \tag{4.31}$$

The plug-in $\widehat{V}(P)$ applies the variance calculation to this empirical table. Equation (4.21) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.32}$$

The observations are independent, so

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.33}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.34}$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.35}$$

To interpret Equation (4.35), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.36}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution, whereas greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'} \widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j} \widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right]. \end{aligned} \tag{4.37}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij} \left\{ \widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl} \widehat{g}_P(k,l) \right\}^2 . \tag{4.38}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P \widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.39}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Combining the two samples' degrees of freedom.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\widehat{\nu}_{\text{expanded}} = \frac{\left\{ \widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q \right\}^2}{\left\{ \widehat{V}(P)/n_P \right\}^2 / \widehat{\nu}_P + \left\{ \widehat{V}(Q)/n_Q \right\}^2 / \widehat{\nu}_Q}}. \tag{4.40}$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\text{expanded}})\right\}, \tag{4.41}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid, so every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. Expanded Welch cannot raise rejection where Wald is already too conservative. Appendix A.5 examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.42}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.43}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.44}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.37) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.45}$$

Combining them with Equation (4.40) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, MI is zero and its first-order change is also zero. The leading changes in MI, the pointwise-MI variance $V(P)$, and the variance sensitivity $\tau^2(P)$ can therefore be compared using a second-order expansion. Appendix A.2 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.46}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.47}$$

This population approximation describes the estimated degrees of freedom when sampling errors are small enough for the first-order calculation to be accurate. Smaller population MI then gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

Very close to independence, sampling noise itself creates apparent MI in the empirical table. The resulting variance estimates need not be close to their population values, so the estimated degrees of freedom need not tend to zero with population MI at a fixed sample size. These local expansions assume fixed table dimensions, positive cell probabilities and the fixed-margin path. Appendix A.4 retains the quadratic sampling terms needed when population MI is very small.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) are exact derivatives for positive population tables and provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.2) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and Expanded Welch's larger cutoff does not account for how the estimated MI difference and its standard error move together. Appendix A.3 gives the first-order covariance between each population's MI and variance estimates. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The experiment evaluates the moment approximations used to derive Expanded Welch and the false-positive rates and power of the resulting test. It first measures how the estimated MI variance behaves across repeated samples from known populations. It then compares four rejection rules using the same bias-corrected MI difference and standard error. This separates errors in the derived moments from the performance of a Student t distribution applied to the shared statistic. The design follows the aims, data-generating mechanisms, targets, methods and performance measures framework of Morris et al. (2019).

5.1 Research questions and statistical targets

For a fixed population P and sample size n_P, expectations and variances below refer to repeated independent multinomial samples. The estimator $\widehat{V}(P)$ estimates the pointwise-MI variance $V(P)$; $\widehat{V}(P)/n_P$ is the estimated contribution to the variance of the MI difference. The sampling variance of $\widehat{V}(P)$ is a different quantity: it describes how that estimated contribution fluctuates across samples.

The first-order calculation uses

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}, \tag{5.1}$$

where $\tau^2(P)$ is the population variance of the variance influence function derived in Chapter 4. Substitution into the moment-matching formula gives the population first-order value

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}. \tag{5.2}$$

The sampling-moment target instead uses the actual mean and variance of the estimator:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.3}$$

The assumptions in Chapter 4 ensure positive population variance quantities for the tested populations. The two degrees-of-freedom values can nevertheless differ at finite sample sizes. Dividing their definitions before applying either approximation gives the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \left\{\frac{V(P)}{\mathrm{E}\,\widehat{V}(P)}\right\}^2 \frac{\mathrm{Var}\{\widehat{V}(P)\}}{\tau^2(P)/n_P}. \tag{5.4}$$

The first factor measures the contribution of the mean approximation; the second measures the contribution of the sampling-variance approximation. A mean estimate above $V(P)$ lowers the first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for Q.

For each population, the experiment estimates the two moments directly and compares the population predictions in Equation (5.1) with them. It also checks the mean plug-in prediction $\mathrm{E}(\widehat{\tau}^2)/n_P$, because the implemented test estimates $\tau^2(P)$ from each sample. Component means, medians and percentile ranges of $\widehat{\nu}_P$ describe the additional effect of substituting empirical probabilities into the first-order formula.

The second question concerns false-positive rates and power relative to Normal Wald and a sample-size comparator. Even accurate denominator moments do not determine the joint distribution of the MI difference and its standard error. The moment checks and rejection comparisons therefore answer distinct parts of the evaluation.

5.2 Four rejection rules with a shared statistic

Every rule uses the bias-corrected difference in Equation (3.7) and the standard error in Equation (3.9). Normal Wald compares their ratio T with the standard normal distribution. Expanded Welch uses its existing estimated degrees of freedom from Chapter 4.

The sample-size comparator uses Hutcheson's assignment of sample size to each component (Hutcheson, 1970), applied to the shared MI variance contributions. It

replaces the component degrees of freedom by n_P and n_Q, giving

$$\widehat{\nu}_{\text{size}} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/n_P + \{\widehat{V}(Q)/n_Q\}^2/n_Q}. \tag{5.5}$$

This is a comparator for the degrees-of-freedom assignment: the numerator and standard error remain the MI quantities from Chapter 3. The component assignment is n_i, whereas ordinary Welch's comparison of sample means uses $n_i - 1$ (Welch, 1947). The experiment asks whether calculating the MI variance sensitivity provides a benefit over the sample-size assignment.

A fourth rule uses a fixed moment-matched reference, estimated from independent simulations of the corresponding equal-MI populations:

$$\nu_{\text{ref}} = \frac{2\{\mathrm{E}(\widehat{\mathrm{SE}}^2)\}^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)}. \tag{5.6}$$

Independence of the population samples gives

$$\mathrm{E}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{E}\,\widehat{V}(P)}{n_P} + \frac{\mathrm{E}\,\widehat{V}(Q)}{n_Q}, \tag{5.7}$$

$$\mathrm{Var}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{Var}\{\widehat{V}(P)\}}{n_P^2} + \frac{\mathrm{Var}\{\widehat{V}(Q)\}}{n_Q^2}. \tag{5.8}$$

Substitution of the component moment targets into the combination formula at these expected variance contributions gives exactly Equation (5.6). Thus the combination identity introduces no further approximation at the level of moments. Appendix B.7 gives the cancellation explicitly. Independent fitting samples estimate these moments by their sample means and sample variances. The Student t distribution remains an approximation to the shared statistic's distribution.

For each finite degrees-of-freedom rule, the nominal two-sided p-value is

$$p = 2\{1 - F_\nu(|T|)\}, \tag{5.9}$$

where F_ν is the distribution function of the Student t distribution and ν is the rule's sample-dependent or fitted value. A valid p-value at most 0.05 produces a rejection.

The moment reference requires known simulation populations and is held fixed within an evaluation setting and across the alternatives belonging to the same null. Expanded Welch versus this reference measures the change in rejection when the sample-dependent degrees of freedom are replaced by that fixed value. The reference can still reject incorrectly when the shared difference is biased, its standard error

has the wrong scale, or their joint distribution is poorly described by the Student t distribution.

5.3 Constructing joint distributions with known MI

The construction changes association while preserving the category frequencies. For one population, draw X with category probabilities $a = (a_1, \ldots, a_k)$. With probability λ, set $Y = X$; otherwise draw Y independently with the same probabilities a. Copying makes matching categories more likely than under independence. Both branches give Y the same marginal probabilities, so changing λ does not change the row or column totals.

An observation in a diagonal cell can arrive through either branch, whereas an off-diagonal observation can arrive only through the independent branch. Combining these probabilities gives

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \le \lambda < 1, \tag{5.10}$$

where the indicator is one when the two categories match and zero otherwise. This construction is a special case of the mixture model described by Bocci et al. (2010). At $\lambda = 0$, the variables are independent and MI is zero. Increasing λ increases MI, so the target MI determines the required copying probability. Appendix B gives the numerical search and shows why the margins remain fixed.

Constructing a 2×2 joint distribution with positive MI.

For P, take $a = (0.7, 0.3)$ and a copying probability of approximately 0.20347. Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, whereas an independent draw gives $Y = 1$ with probability 0.7. Thus

$$\begin{aligned} p_{11} &\simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273, \\ p_{12} &\simeq 0.7(1 - 0.20347)(0.3) = 0.16727. \end{aligned}$$

The second equation has no copying term because copying cannot produce $X = 1, Y = 2$. The remaining cells follow the same rule. For Q, use $a = (0.8, 0.2)$ and a copying probability of approximately 0.26091. These choices give $I(P) = 0.02$ and

$I(Q) = 0.03$ nats:

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.11}$$

Both joint distributions have positive MI, and the true difference used to assess power has magnitude 0.01 nats.

Table sizes and margin regimes.

The copying construction is used for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four, so at a fixed sample size the observations are distributed over more possible cells.

Three margin regimes vary the category frequencies: uniform frequencies in both populations, the same uneven frequencies in both, or different uneven frequencies between them. For uneven margins, the first category has probability ρ, and the remaining mass is shared equally:

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.12}$$

Each population uses the same vector for its row and column margins. Table 5.1 specifies the vectors for P and Q. The copying probabilities for P and Q are chosen separately to achieve the targets $I(P)$ and $I(Q)$.

Table 5.1: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$. In contrast, the different-skew regime gives different tables with equal MI because the category frequencies differ between populations. Comparing the regimes therefore tests false-positive behaviour under both identical and distinct population tables.

Same skew provides a matched baseline for different skew in all three studies and both supporting checks. With equal MI and equal sample sizes, the same-skew samples have identical distributions, so their expected MI estimation biases cancel in the difference. Different-skew populations can retain unequal biases despite having

equal MI. Under an MI difference or unequal sample sizes, matching margins alone does not ensure that biases cancel. The two regimes therefore distinguish failures that also occur with matching category frequencies from those observed when the frequencies differ.

Increasing table size under the uneven-margin rule spreads the remaining mass over more categories, changing both the number of cells and their probabilities. This comparison shows what happens under a common margin rule as the table grows. Uniform versus same skew instead compares even and uneven category frequencies at a fixed size; different skew allows the two populations' frequencies to differ as well.

5.4 Settings and controlled comparisons

The equal-MI comparisons include identical joint tables under uniform and same-skew margins, and distinct joint tables under different-skew margins. This distinction is necessary because the null concerns equal MI rather than equal cell probabilities. The experiment uses six comparisons, with overlapping settings sampled only once:

Table 5.2: Experimental comparisons and their statistical purposes.

Comparison	Settings	Main purpose
Broad equal MI	120	Approximation error, false positives and validity.
Population-MI sweep	63	Behaviour near independence and across MI.
Focused power	27	Nominal and null-calibrated power at A–C.
Broader power	162	Table-size, margin and direction comparisons.
Construction	32	Matched copying and four-cell populations.
Allocation	12	Fixed total sample size with both allocation directions.

The broad equal-MI study crosses four table sizes, three margin regimes and ten sample sizes at common MI 0.02 nats. Its 120 settings assess false-positive rates and approximation accuracy across population structure and sampling density. The MI sweep uses 4×4 tables, all three margin regimes, seven positive MI targets and sample sizes 100, 1,000 and 10,000. After removing shared settings, these two studies contain 174 distinct settings. Across all six comparisons there are 366 distinct evaluation settings and 186 distinct equal-MI fitting settings. Appendix B lists the complete grids.

The MI sweep compares population and plug-in predictions for the sampling variance of $\widehat{V}$, and checks the near-independence formulas in Appendix A.4. Those

formulas predict means for each population separately. Observed means are therefore compared with predicted means; medians describe the additional effect of skewed sampling distributions. Increasing n while retaining the same population distinguishes sampling effects from changes in population MI. A large noncentrality parameter alone is not taken as a failure condition.

The scaled mean error $n\{\mathrm{E}\,\widehat{V} - V\}$ is also compared with $d = (k-1)^2$. The near-independence prediction suggests a contribution close to d/n, whereas away from independence the second-order coefficient can depend on the population table. The experiment measures that dependence without assuming a universal correction.

Table 5.3: Focus settings for the power comparison; common null MI is 0.02 nats.

Setting	Margins	$n_P = n_Q$	Size
A	Uniform	20	4×4
B	Different skew	100	8×8
C	Different skew	10,000	4×4

Settings A and B were selected from earlier simulations because Expanded Welch brought their false-positive rates closer to 5%. Independent simulations estimate their performance separately from those earlier samples. Setting C provides a larger-sample comparison. The broad study supplies the wider evidence needed to interpret these selected cases. Four matched failure settings retain 16×16 populations at $n = 1{,}000$ and 10,000 under both skew regimes.

For each focus null, estimate $\sigma_0 = \mathrm{SD}_0(\widehat{\Delta}_{\mathrm{BC}})$ from an independent fitting batch, centring at its sampling mean. Construct alternatives with

$$I(P) = 0.02, \qquad I(Q) = 0.02 + h\sigma_0, \qquad h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}. \tag{5.13}$$

The signed difference is $\Delta = -h\sigma_0$. Report each difference in nats as well as null standard deviations, and hold the fitted population tables fixed during evaluation.

The broader power study retains the existing differences from 0 to 0.2 nats at $n = 1{,}000$ and 10,000. It includes matched same-skew and different-skew comparisons at every size, with uniform 4×4 populations as an additional comparison. The four-cell check matches MI, margins and sample sizes to the copying construction. The allocation check holds the total sample size at 2,000 and compares $(1000, 1000)$, $(400, 1600)$ and $(1600, 400)$ for null and 0.01-nat alternatives under both skew regimes. It assesses the whole procedure under imbalance, including bias and estimated variance weights.

A separate, earlier bias-correction investigation is retained in Appendix C.7. It compares an analytic correction with a delete-one jackknife on 20,000 pairs per setting and does not add an arm to the four-rule experiment.

5.5 Independent fitting, evaluation and calibration

Every distinct evaluation setting uses 100,000 independently sampled pairs of count tables. Population samples within a pair are independent; repetitions are also independent. A separate batch of 100,000 pairs is drawn at each of the 186 null settings to fit the component moments and fixed reference. All four rules receive the same evaluation pairs. Repeated settings share one evaluation batch across displays. Separate random streams supply the two populations, each setting, and the fitting and evaluation batches.

For each alternative, the reference is fitted under its corresponding null with the same table size, margins, construction and sample allocation. The baseline MI in both populations is 0.02 nats. Thus changes in power are assessed against a fixed null reference rather than one refitted at each alternative.

For the focused power comparison, the fitting batch also supplies empirical null calibration thresholds. Each sample-dependent rule rejects when its p-value lies strictly below its fitted fifth-percentile threshold. Ties at the boundary are excluded, giving a fitting rejection rate no greater than 5%. Normal Wald and the fixed reference use the same empirical cutoff for $|T|$, since both order the samples by that quantity. Their calibrated decisions consequently coincide. These thresholds and the reference degrees of freedom remain fixed throughout each alternative curve. The independent null evaluation batch measures the achieved false-positive rate.

Nominal power uses each rule's stated 5% cutoff. Calibrated power compares the rules after their null rejection rates have been matched empirically. The latter measures discrimination after adjustment for null calibration; the fitted thresholds use known populations and are not proposed as an additional data-analysis method.

5.6 Performance measures and simulation uncertainty

Primary rejection rates divide valid rejections by every attempted pair. Invalid outputs are recorded separately and produce no rejection. Method-valid and

common-valid rates retain their stated denominators. Moment calculations include finite zero variance estimates and finite estimates from invalid tests. Component degrees-of-freedom summaries use positive finite values and report their included counts. If exclusions occur, those conditional summaries do not estimate an unconditional mean.

For each population, report $\mathrm{E}\,\widehat{V}/V$, predicted over observed sampling variance, first-order over moment degrees of freedom, and the mean, median and fifth–95th percentile range of estimated component degrees of freedom. These percentile ranges describe sampling spread. They are distinct from confidence intervals for a summary.

For the shared difference, report bias divided by observed SD and mean squared standard error divided by observed sampling variance. Correlations, skewness and the two rejection tails provide additional diagnostics at the focus and failure cases. Under alternatives, classify rejections by whether $\widehat{\Delta}_{\mathrm{BC}}\Delta$ is positive or negative, retaining both in the two-sided rejection rate.

Wilson 95% intervals describe uncertainty in rejection rates (Wilson, 1927). Paired differences subtract the comparator's rejection indicator from Expanded Welch's indicator on each sampled pair. Their Monte Carlo standard errors use the observed variance of these paired differences. At a true rate of 5%, 100,000 repetitions give an approximate Monte Carlo standard error of 0.069 percentage points. Nominal reference intervals condition on the fitted degrees of freedom; its fitting uncertainty is reported separately.

Moment-ratio uncertainty uses the delta method applied to the joint sample moments, as derived in Appendix B.8. Focused calibrated-power intervals use 1,000 bootstrap repetitions of the fitting and evaluation outputs, preserving paired method outcomes and the fitting batch shared across each curve. The population tables and differences in nats remain fixed. A power difference within one percentage point is the stated practical tolerance, assessed with pointwise uncertainty intervals rather than a nonsignificant test alone. Intervals quantify simulation uncertainty for the specified populations; they do not provide simultaneous coverage over the design.

Chapter 6

Results

The experiment contains 366 evaluation settings with 100,000 sampled pairs in each. Independent 100,000-pair batches at 186 equal-MI settings fit the moment references and calibration thresholds. This chapter first checks the component approximations, then compares the four rejection rules. All displayed nominal rates divide valid rejections by all attempts. Brackets around rates and shading in the rejection plots give pointwise 95% intervals. Reference-rule intervals condition on the independently fitted degrees of freedom. Complete setting-level rates, denominators and paired contrasts are supplied in the experiment supplement.

6.1 Mean and sampling variance of the estimated MI variance

The mean approximation and sampling-variance approximation have different errors. Figure 6.1 separates them for different-skew populations at common MI 0.02 nats. A ratio of one indicates agreement. Solid curves use the population quantities; dashed curves in the middle row use the mean plug-in sampling-variance prediction. Shading gives approximate 95% Monte Carlo intervals for the population-prediction ratios. The component mean remains inflated after the prediction of its sampling variance has become reasonably accurate. At $n = 1{,}000$, the population sampling-variance prediction divided by the observed value ranges from 0.710 to 0.996 across all sizes and margins, while the mean estimate divided by V ranges from 1.023 to 4.571. The latter discrepancy is particularly large for the larger tables.

At $n = 50{,}000$, the sampling-variance ratios across the same broad grid lie between 0.974 and 1.009. Mean ratios range from 1.000 to 1.091, so small residual mean errors can still affect the degrees of freedom. The first-order/moment degrees-

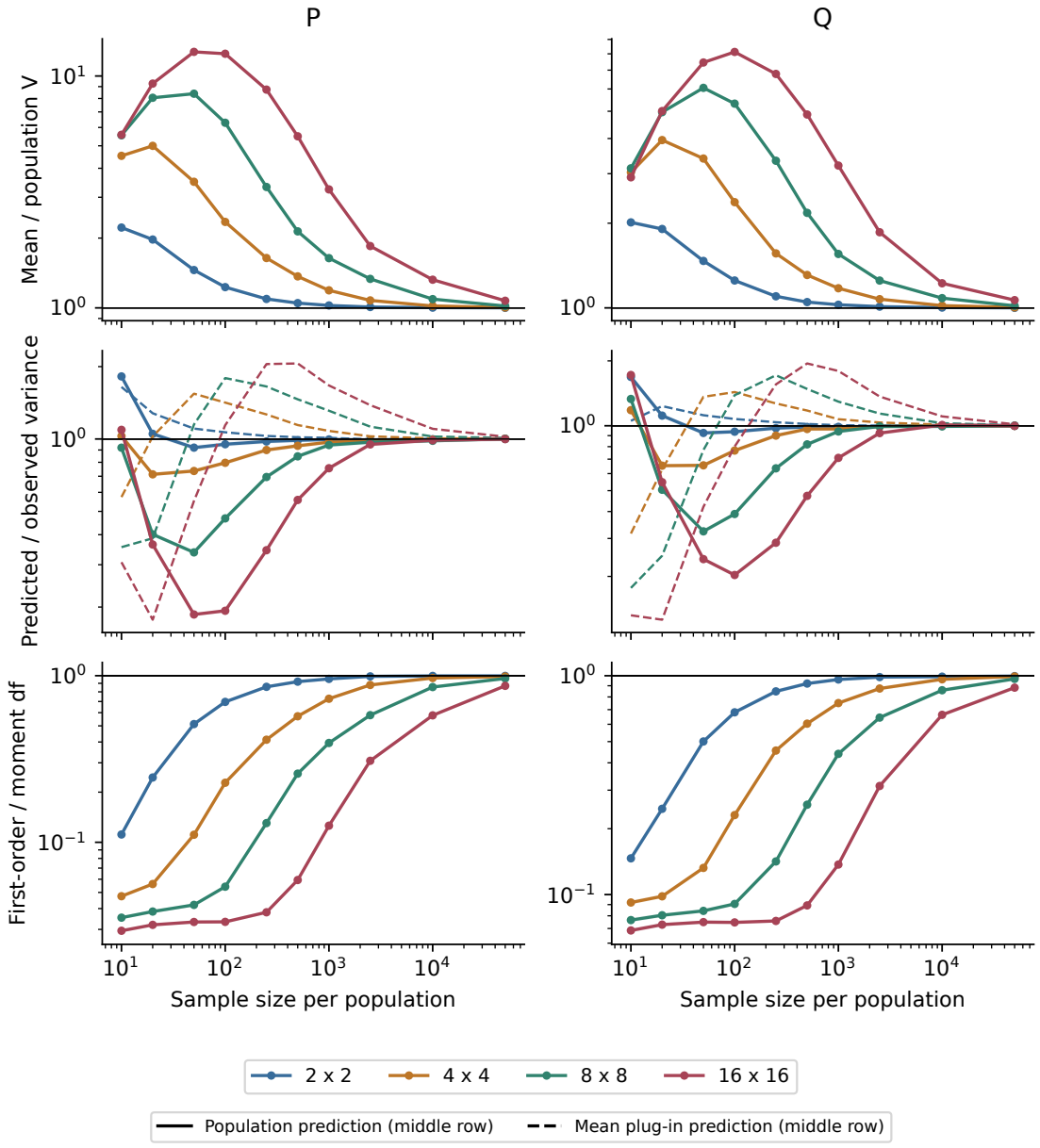


Figure 6.1: Component approximation ratios under different-skew margins.

of-freedom ratios range from 0.862 to 1.006. Thus increasing sample size improves both approximations, but their rates of improvement differ. Appendix C provides the uniform and same-skew versions of the figure.

The decomposition in Equation (5.4) explains this difference. At 4×4, $n = 1{,}000$ and MI 0.02, the sampling-variance ratios are 0.970 for P and 0.967 for Q. Their mean ratios are 1.190 and 1.174. Consequently, the population first-order degrees of freedom, 15.30 and 13.61, fall below the moment targets, 21.03 and 18.14. Most of that difference comes from squaring the mean-ratio contribution, rather than from an inaccurate sampling-variance prediction. Table 6.1 gives the component values.

Table 6.1: Component moment and degrees-of-freedom comparisons at 4×4, $n = 1{,}000$, different skew.

MI, pop.	Mean ratio	Pop. var.	Plug-in var.	Mean df	Moment df	FO df
0.0001, P	44.939	0.047	1.965	4.86	9.12	0.10
0.0001, Q	41.996	0.052	2.142	4.44	8.63	0.09
0.02, P	1.190	0.970	1.080	19.55	21.03	15.30
0.02, Q	1.174	0.967	1.072	17.00	18.14	13.61
0.05, P	1.066	1.008	1.013	42.56	43.07	37.63
0.05, Q	1.057	1.031	1.029	37.59	38.61	33.50

In the table, mean ratio is $\mathrm{E}\,\widehat{V}/V$; the next two columns divide the population and mean plug-in sampling-variance predictions by the observed sampling variance of $\widehat{V}$. Mean df averages the positive finite component estimates; all 100,000 estimates are included in each shown row. Moment df uses the actual sampling moments, while FO df uses the population first-order formula.

6.2 Near independence and the Appendix A.4 predictions

At MI 0.0001 and $n = 1{,}000$, the quadratic sampling contribution is large relative to the population first-order variance. The observed values of $n\,\mathrm{E}\,\widehat{V}$ are 9.158 for P and 8.665 for Q, compared with the Appendix A.4 prediction 9.2. Mean estimated component degrees of freedom are 4.86 and 4.44, compared with 4.6. The moment values are 9.12 and 8.63, compared with approximately 9.00. These are component means and moment targets; a median combined degree of freedom would answer a different question.

Figure 6.2 shows how that agreement changes across MI. The shaded band gives the fifth–95th percentile range of estimated component degrees of freedom,

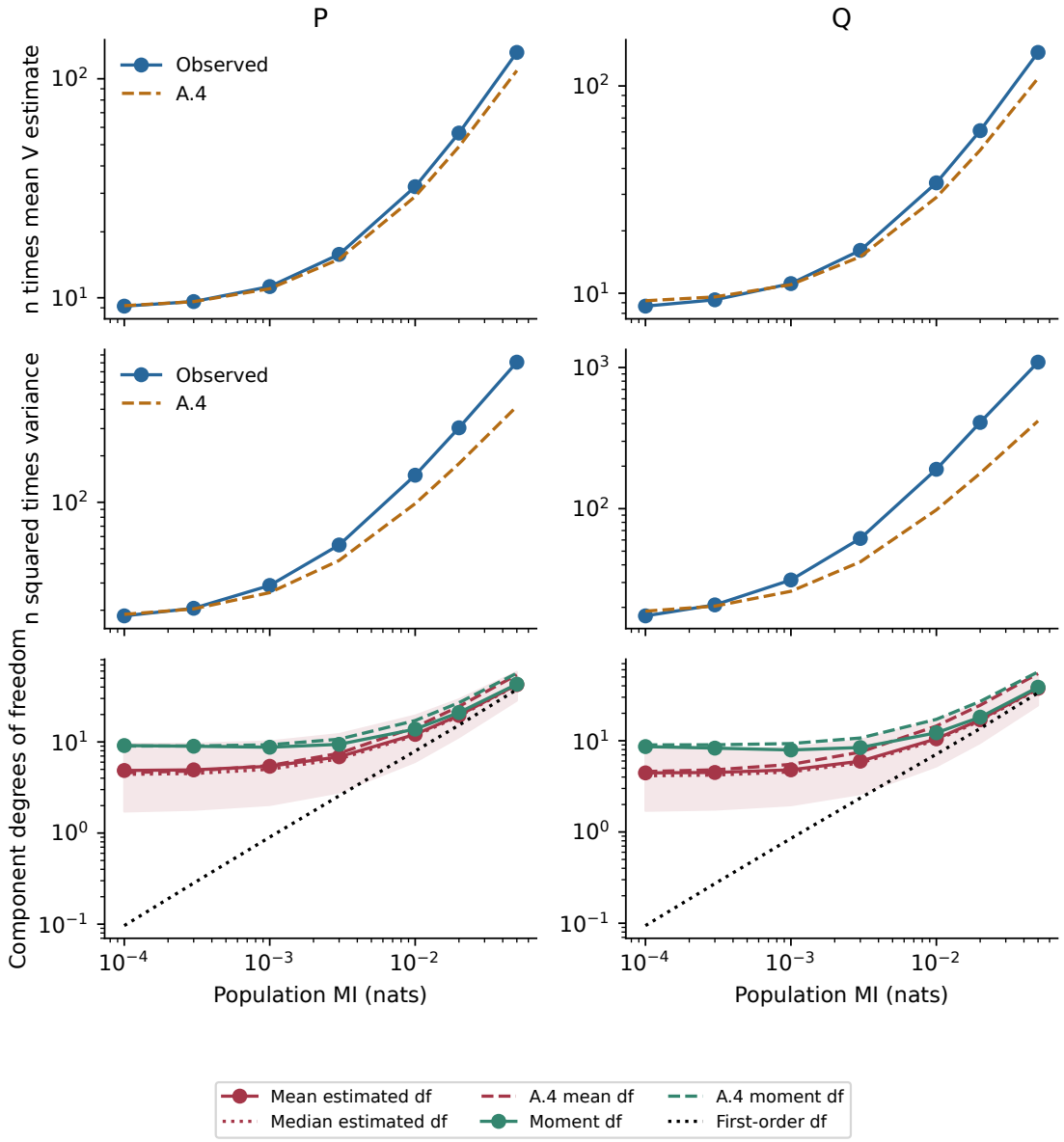


Figure 6.2: Observed component moments and Appendix A.4 predictions across MI.

with their mean and median shown separately. The first-order prediction for the sampling variance is only about 5% of the observed value at the smallest MI target. Meanwhile, the mean plug-in prediction is 1.97 times the observed value for P and 2.14 times for Q. The local calculation explains the scale of these errors: the quadratic sampling contribution remains substantial even though the population first-order degrees of freedom are below 0.1. At MI 0.02, the population sampling-variance approximation is much more accurate, while the mean error still lowers the first-order degrees of freedom relative to the moment target. The mean plug-in variance predictions remain about 7–8% above the observed values. Thus the accuracy of the population variance-sensitivity formula does not eliminate the error from estimating that sensitivity in each sample.

The scaled mean error varies around $d = 9$. Across the three margin regimes and seven MI targets, its observed range is 5.55–8.45 at $n = 100$, 7.50–9.39 at $n = 1{,}000$, and 7.60–10.19 at $n = 10{,}000$. Figure 6.3 displays the changes with population MI and margins, with 95% Monte Carlo intervals. The coefficient near nine explains the persistent upward shift in the component mean for these 4×4 populations. Its variation with the population table and sample size determines where a fixed d/n correction would leave residual error. Companion comparisons at $n = 100$ and 10,000 are in the appendix.

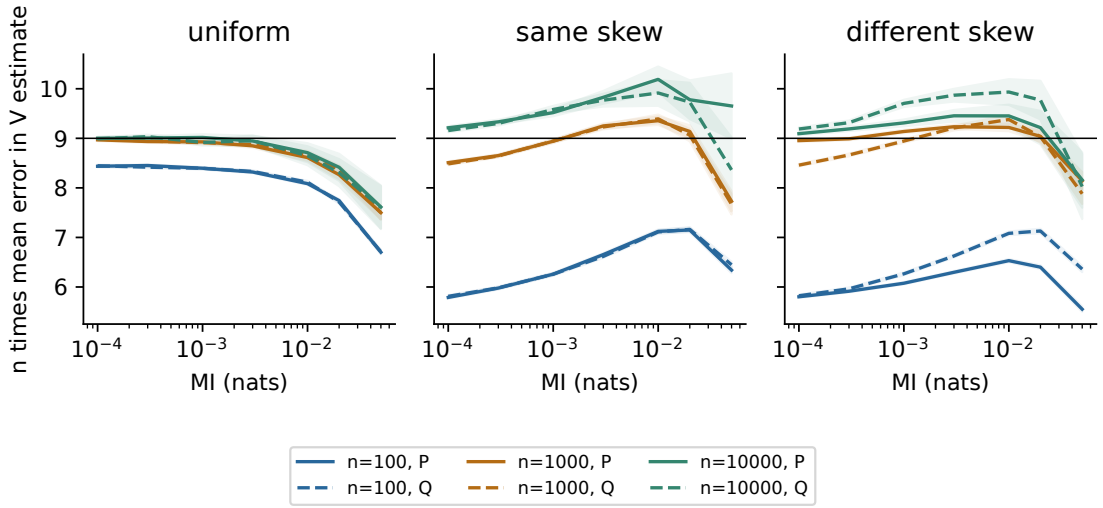


Figure 6.3: Scaled mean error of the variance estimator across population MI.

6.3 False-positive rates across the broad study

Figures 6.4 and 6.5 compare the four rules at common MI 0.02 nats. Expanded Welch reduces false positives in some small-sample settings and lowers already-conservative rates in others. The sample-size comparator often remains close to

Normal Wald once sample sizes are moderate. The moment reference also varies in its calibration, showing that accurate denominator moments alone do not provide a correctly calibrated Student t reference for every population.

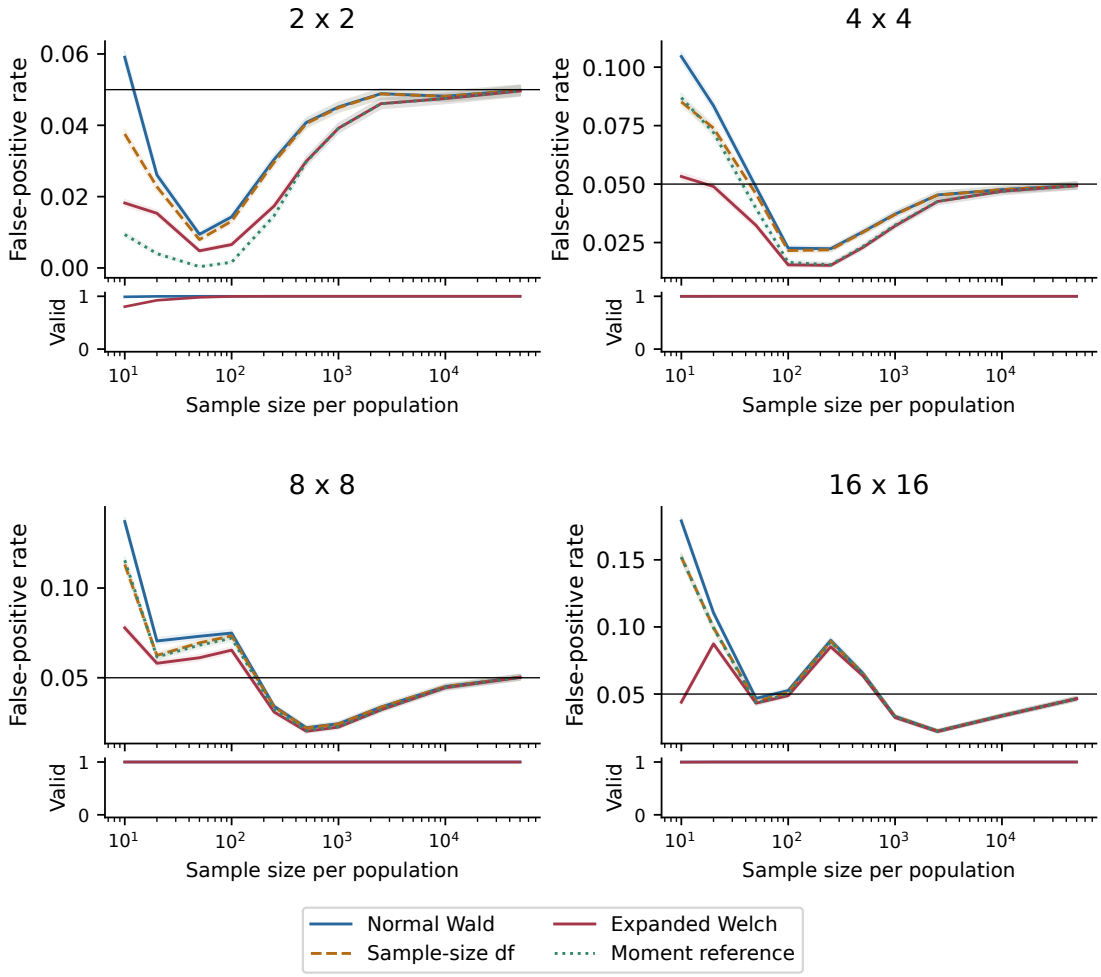


Figure 6.4: False-positive rates under uniform margins.

The lower panels show the fraction of valid outputs for Normal Wald and Expanded Welch. The sample-size and fixed-reference rules share Wald's validity in these settings. Validity is lowest in the smallest samples: across the 120 settings, the minimum valid fractions are 0.958 for Normal Wald, the sample-size comparator and the fixed reference, and 0.631 for Expanded Welch. Their conditional rates and included counts are reported separately in the supplement.

Across the broad grid, the counts of rates between 4% and 6% are 43 for Wald, 41 for the sample-size comparator, 41 for Expanded Welch and 35 for the fixed reference. Expanded Welch's small-sample improvements therefore do not extend to uniformly better calibration across the grid. The different-skew large-table failure and the conservative same-skew cases explain why a lower rejection rate can improve or worsen agreement with the 5% target.

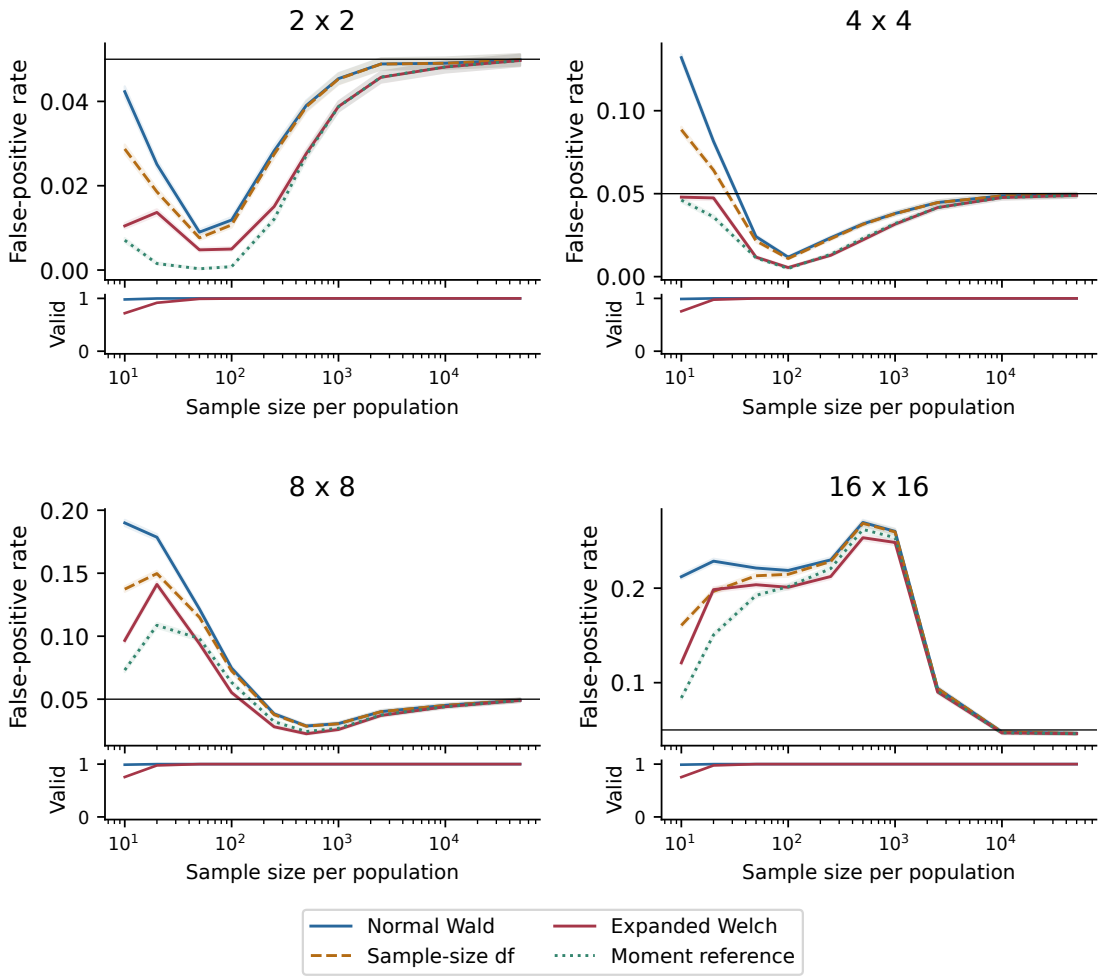


Figure 6.5: False-positive rates under different-skew margins.

6.4 Focus settings and shared estimation failures

Table 6.2 compares the focus settings with the matched large-table cases. All four rules are valid on every sampled pair in these seven settings. At A, Expanded Welch lowers rejection from 0.08352 for Wald and 0.07394 for the sample-size comparator to 0.04897. The fixed moment reference gives 0.07202. At B, the corresponding rates are 0.07466, 0.07246, 0.05523 and 0.06331. The B estimate therefore remains above 5%, although it is closer than the comparator rates.

Table 6.2: Null rejection rates with pointwise 95% intervals.

Setting	Normal Wald	Sample-size df
A	0.0835 [0.0818, 0.0853]	0.0739 [0.0723, 0.0756]
B	0.0747 [0.0730, 0.0763]	0.0725 [0.0709, 0.0741]
C	0.0486 [0.0473, 0.0500]	0.0486 [0.0473, 0.0500]
16, same, 1,000	0.0060 [0.0056, 0.0065]	0.0060 [0.0055, 0.0065]
16, same, 10,000	0.0438 [0.0426, 0.0451]	0.0438 [0.0425, 0.0451]
16, different, 1,000	0.2601 [0.2574, 0.2628]	0.2595 [0.2568, 0.2622]
16, different, 10,000	0.0478 [0.0465, 0.0491]	0.0478 [0.0464, 0.0491]

Setting	Expanded Welch	Moment reference
A	0.0490 [0.0476, 0.0503]	0.0720 [0.0704, 0.0736]
B	0.0552 [0.0538, 0.0567]	0.0633 [0.0618, 0.0648]
C	0.0477 [0.0464, 0.0490]	0.0477 [0.0464, 0.0490]
16, same, 1,000	0.0053 [0.0048, 0.0057]	0.0056 [0.0052, 0.0061]
16, same, 10,000	0.0426 [0.0414, 0.0439]	0.0427 [0.0415, 0.0440]
16, different, 1,000	0.2484 [0.2457, 0.2510]	0.2535 [0.2508, 0.2562]
16, different, 10,000	0.0468 [0.0455, 0.0482]	0.0469 [0.0457, 0.0483]

Table 6.3: Shared-estimator diagnostics and combined degrees of freedom.

Setting	Bias/SD	Var. ratio	Corr.	Skew	Median df	Ref. df
A	0.003	0.827	0.006	0.002	10.61	29.70
B	0.870	1.535	-0.177	-0.052	21.12	33.34
C	0.000	1.010	-0.091	-0.013	293.23	293.98
16, same, 1,000	-0.003	1.929	0.005	0.008	78.87	142.90
16, same, 10,000	0.001	1.047	-0.006	0.012	265.85	294.14
16, different, 1,000	2.058	1.938	-0.129	-0.008	88.51	155.70
16, different, 10,000	-0.229	1.064	-0.090	-0.013	299.21	329.92

In Table 6.3, bias/SD divides difference bias by its observed sampling SD; variance ratio divides mean squared standard error by observed difference variance.

Corr. is the correlation between the difference and its standard error, and skew describes the difference estimates. These diagnostics describe the shared statistic without identifying separate causal contributions.

At A, difference bias is negligible, but the estimated variance is only 0.827 of the observed variance. Its median estimated combined degrees of freedom are 10.61, compared with the fixed moment reference 29.70. At B, the bias is 0.870 observed SDs and the variance ratio is 1.535. Its median degrees of freedom are 21.12, compared with reference 33.34. The lower median degrees of freedom at A and B give larger cutoffs than the fixed reference on a typical sample, while the sample dependence of those cutoffs also changes which pairs are rejected. Thus the two small-sample improvements occur alongside different estimation errors. At C, bias is negligible, the variance ratio is 1.010, and median and reference degrees of freedom are both about 293–294.

The 16×16, different-skew case at $n = 1{,}000$ remains a clear failure for all four rules. The difference bias is 0.02985 nats, about 2.058 observed SDs. Rejection ranges from 0.24836 to 0.26011. The reference rate is 0.25348 despite its independently fitted moments. Under the equal-MI null, the estimated difference is centred at

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx 0 + 0.02985 = 0.02985 \text{ nats}. \tag{6.1}$$

This shift is large relative to the observed SD of approximately 0.01450 nats. It moves most rejections into the positive tail even though the population difference is zero. With same skew, difference bias is negligible and the variance ratio is 1.929; all four rates instead lie between 0.00525 and 0.00604. The matched populations separate the large difference bias from conservative behaviour that occurs when the biases cancel. At $n = 10{,}000$, both skew regimes give much smaller estimation errors and rates closer to 5%.

6.5 Power at the selected settings

6.5.1 Nominal power

Figure 6.6 measures power along the fixed paths in Equation (5.13). Expanded Welch has lower nominal power than the sample-size comparator at A and B. At two null SDs, the A difference is 0.30993 nats: Expanded Welch rejects 0.35889 of pairs, compared with 0.38875 for the sample-size comparator and 0.41017 for Wald. At B, two null SDs is 0.13028 nats, with rates 0.06046, 0.06546 and 0.06773,

respectively. These comparisons include the different null rejection rates shown at zero.

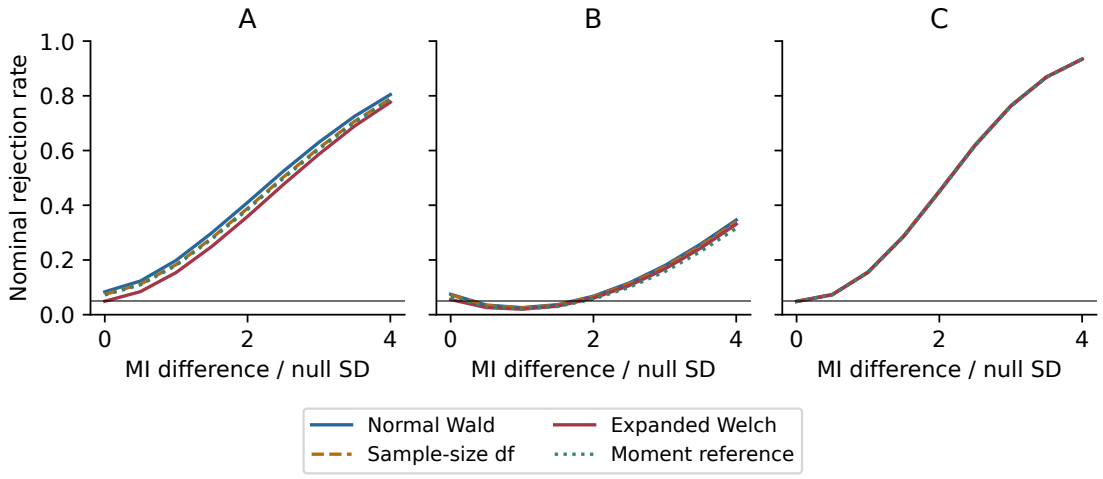


Figure 6.6: Nominal power across differences measured in fitted null standard deviations.

B also shows a falling-then-rising rejection curve. Expanded Welch's rate falls from 0.05523 under the null to 0.01993 at one null SD, before rising to 0.33075 at four SDs. The true signed difference becomes negative as $I(Q)$ increases, initially opposing the positive bias observed under the null. The larger rejection cutoff does not remove this distortion of the shared statistic. Direction counts and the broader large-table curves provide supporting evidence in the appendix.

At C, nominal power differences are small. At two null SDs, or 0.00635 nats, Wald gives 0.45206 and Expanded Welch gives 0.44902. The paired loss is 0.304 percentage points, with 95% interval 0.270–0.338. Table 6.5 reports paired comparisons at two null SDs for all three focus settings. These intervals use the same sampled pairs, so they measure the difference directly. The calibrated comparison assesses how that difference changes after matching null rejection.

6.5.2 Power after empirical null calibration

Figure 6.7 and Table 6.4 use independent null fitting to compare rules at approximately 5% rejection. Null evaluation rates range from 0.04926 to 0.04994 at A, 0.04961 to 0.04977 at B, and 0.04976 to 0.04981 at C. Wald and the fixed reference have identical decisions after calibration.

Expanded Welch has higher calibrated power at A. At two null SDs its rate is 0.36145, compared with 0.32565 for Wald; the paired gain is 3.580 percentage points, with bootstrap 95% interval 3.295–3.787. At three SDs, the gain is 4.496 points, with interval 4.230–4.766. The sample-size comparator remains close to calibrated

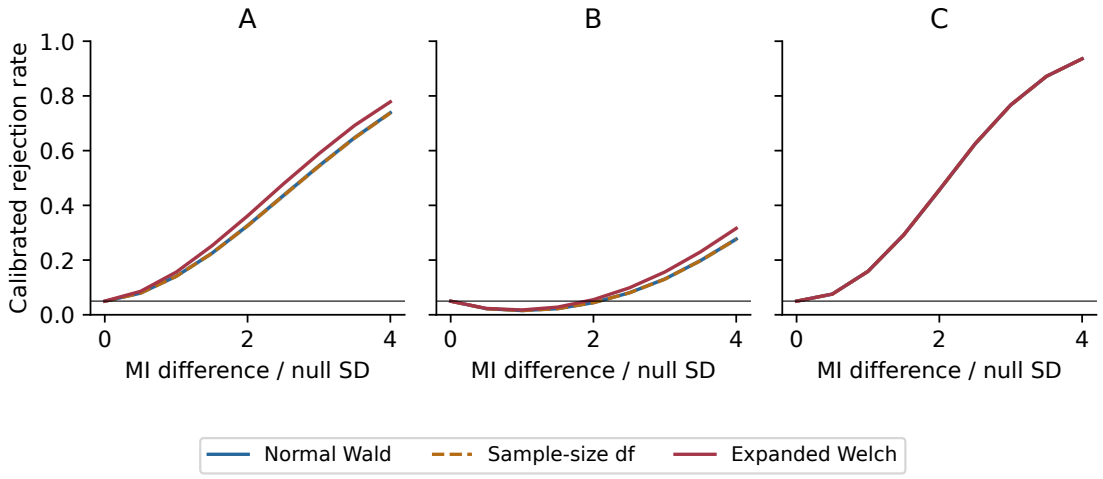


Figure 6.7: Focused power after independent null calibration; bands show bootstrap uncertainty.

Table 6.4: Calibrated rejection rates and actual MI differences.

Setting: SD	Nats	Wald/ref.	Size df	Expanded
A: 0	0.00000	0.0493	0.0493	0.0499
A: 1	0.15497	0.1407	0.1403	0.1559
A: 2	0.30993	0.3256	0.3252	0.3614
A: 3	0.46490	0.5441	0.5433	0.5891
A: 4	0.61987	0.7384	0.7377	0.7784
B: 0	0.00000	0.0498	0.0498	0.0496
B: 1	0.06514	0.0154	0.0154	0.0177
B: 2	0.13028	0.0442	0.0441	0.0554
B: 3	0.19541	0.1307	0.1303	0.1564
B: 4	0.26055	0.2766	0.2761	0.3159
C: 0	0.00000	0.0498	0.0498	0.0498
C: 1	0.00317	0.1585	0.1585	0.1588
C: 2	0.00635	0.4561	0.4561	0.4566
C: 3	0.00952	0.7668	0.7668	0.7672
C: 4	0.01270	0.9356	0.9356	0.9358

Wald. These results show that the sample-dependent MI degrees of freedom can improve the ordering of evidence along this population path, beyond reducing the nominal false-positive rate.

B also gives higher calibrated power for Expanded Welch, although its rejection curve remains distorted. At two SDs, rates are 0.05536 for Expanded Welch and 0.04424 for Wald, a gain of 1.112 percentage points with interval 1.028–1.194. At four SDs, the gain is 3.924 points with interval 3.776–4.151. The initial decline remains after calibration: matching the null rate does not correct the bias-related power failure.

At C, the calibrated differences are practically small. At two SDs the rates are 0.45655 and 0.45611, a gain of 0.044 percentage points with interval 0.013–0.071. The selected one-, three- and four-SD comparisons also have intervals wholly within the stated one-percentage-point tolerance. These are pointwise comparisons for the specified populations. They do not imply a power advantage across untested association patterns.

Table 6.5: Expanded Welch minus each comparator at two null SDs, in percentage points.

Setting, comparator	Nominal	95% interval	Calibrated	95% interval
A, Wald	-5.128	[-5.265, -4.991]	+3.580	[+3.295, +3.787]
A, size df	-2.986	[-3.093, -2.879]	+3.629	[+3.394, +3.865]
B, Wald	-0.727	[-0.780, -0.674]	+1.112	[+1.028, +1.194]
B, size df	-0.500	[-0.544, -0.456]	+1.123	[+1.043, +1.210]
C, Wald	-0.304	[-0.338, -0.270]	+0.044	[+0.013, +0.071]
C, size df	-0.296	[-0.330, -0.262]	+0.044	[+0.013, +0.071]

6.6 Supporting construction and allocation comparisons

6.6.1 Changing the association construction

The four-cell comparison retains the population MI, margins and sample sizes while concentrating dependence in the first two rows and columns. It reproduces the main sparse-table failure. At 16×16, different skew and $n = 1{,}000$, the difference bias is 0.03155 nats, about 2.023 observed SDs. Wald rejects 0.30958 of pairs and Expanded Welch rejects 0.29719; the fixed reference gives 0.30129. The copying rates at the matched setting are 0.26011, 0.24836 and 0.25348. The size of the error

changes with the construction, while severe over-rejection remains under all four rules.

With same-skew margins at the same size and sample count, the four-cell null bias is negligible and the variance ratio is 1.659. Its rates are 0.01126 for Wald and 0.01001 for Expanded Welch. As with copying, matching category frequencies cancels the expected difference bias but leaves conservative rejection. At $n = 10{,}000$, the four-cell variance ratios are 1.038 under different skew and 1.022 under same skew, much closer to one. Table C.3 gives the matched null comparisons, with alternative results in the supplement.

6.6.2 Changing the allocation of a fixed sample total

The allocation comparison retains each 4×4 population pair and distributes 2,000 observations as $(1000, 1000)$, $(400, 1600)$ or $(1600, 400)$. Under the different-skew null, Wald's rates are 0.03808, 0.04500 and 0.04524. Expanded Welch gives 0.03162, 0.02927 and 0.02506. Its median degrees of freedom fall from 35.63 under balance to 14.38 and 11.00 under imbalance, increasing the cutoff and lowering already-conservative rejection.

At the 0.01-nat alternative, allocating only 400 observations to Q reduces Wald power from 0.11483 under balance to 0.05316, and Expanded Welch power from 0.10260 to 0.03423. The observed SD rises from 0.01173 to 0.01611 nats. Giving Q 1,600 observations instead gives Wald power 0.13236 and Expanded Welch power 0.10375. Thus the direction of imbalance matters because the populations contribute different amounts of uncertainty and residual bias.

The same-skew null has identical population tables, but unequal sample sizes leave residual difference biases of approximately $+0.00129$ and -0.00128 nats under the two reversed allocations. These signs reverse with the sample sizes because the MI estimation bias depends on sample size after the leading correction has been applied. Table C.4 reports all three allocations under both nulls and alternatives. The comparison assesses the full method under imbalance; the identity for combining independent component moments remains exact.

The earlier check in Appendix C.7 replaces Wald's analytic MI bias correction with a delete-one jackknife, retaining the existing standard error and standard normal distribution cutoff. Its 20,000-pair samples provide a separate investigation of bias correction, rather than an additional rejection rule. It improves the estimated ordering in the sparse different-skew alternative while leaving conservative null rejection.

6.7 Application to online shopping data

The Online Shoppers Purchasing Intention dataset provides a practical comparison of month–purchase MI between new and returning visitors (Sakar and Kastro, 2018). A larger MI means that the month provides more information about whether a session ends in a purchase. A second comparison uses the weekend indicator instead of month to examine whether its MI with purchasing differs between the same visitor groups.

The dataset contains 12,330 sessions, with each session belonging to a different user according to the source documentation (Sakar and Kastro, 2018). The analysis retains all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable: true indicates a purchase and false indicates no purchase. The tests treat the two disjoint visitor groups as independent samples and the sessions within each group as independent observations.

Each group forms a 10×2 month–purchase count table. The rows are the ten months recorded in the dataset: February, March, May, June, July, August, September, October, November and December. The columns are no purchase and purchase, and each session adds one to its corresponding cell. Both groups use these same categories, with no smoothing or removal of zero cells. All ten months occur in both groups, although the new-visitor February row contains only one session and no purchase. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows; all four cells are observed.

Both methods use the bias-corrected MI estimates in Table 6.6. For the month tables, the leading bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each plug-in MI estimate (Miller, 1955; Moddemeijer, 1989); for the weekend tables, it subtracts $1/(2n)$. Here n is the number of sessions in the relevant visitor group. The estimated difference is new-visitor MI minus returning-visitor MI for the same predictor–purchase pair. Normal Wald compares this difference divided by its standard error with the standard normal distribution; Expanded Welch uses the same statistic with a Student t distribution and estimated degrees of freedom. Both tests are two-sided at the 5% level.

Table 6.6: MI estimates and two-sample test results for online shopping sessions.

	Corrected MI (nats)			p-value	
Predictor	New	Returning	Expanded Welch degrees of freedom	Normal Wald	Expanded Welch
Month	0.004316	0.017880	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	0.000664	0.000727	2.18	0.9554	0.9601

Month has higher estimated MI with purchasing among returning visitors. The corrected difference is -0.013563 nats, with a standard error of 0.003333 nats, giving

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.2}$$

With 26.79 estimated degrees of freedom, the Student t distribution gives a larger p-value than the standard normal distribution, but both tests reject equal month–purchase MI for new and returning visitors. This application was selected from an initial exploration of six comparisons across three public datasets. Accounting for all six comparisons with Holm's multiple-testing adjustment (Holm, 1979) gives month p-values of 0.000189 for Normal Wald and 0.00149 for Expanded Welch, so the conclusion remains unchanged. Table C.7 lists all six comparisons and their raw and adjusted p-values.

The weekend comparison gives a different result: the corrected weekend–purchase MI estimates for new and returning visitors are both approximately 0.0007 nats. Their difference, -0.000063 nats, is small relative to the standard error of 0.001124 nats, giving $T \approx -0.056$. Although Expanded Welch estimates only 2.18 degrees of freedom, both p-values remain close to one. Neither test provides evidence that weekend–purchase MI differs between the visitor groups, but this does not establish that their population MI values are equal. The small statistic, rather than the degrees of freedom alone, explains the agreement between the methods here.

Both methods reach the same decisions in this application, although Expanded Welch returns larger p-values. Its degrees of freedom depend on how much the estimated MI variances fluctuate relative to their mean values, so large session counts alone need not make its p-values close to Normal Wald's.

Chapter 7

Implications and Limitations

7.1 Accuracy of the variance approximation

The variance-sensitivity derivation describes how the estimated MI variance changes between samples. Its accuracy and the accuracy of the mean approximation contribute differently to the degrees of freedom. At 4×4, MI 0.02 and $n = 1{,}000$, the population sampling-variance predictions are within about 3–4% of the observed values, while the mean estimates exceed the population values by about 17–19%. Equation (5.4) shows how squaring the mean-ratio contribution accounts for most of the resulting difference between the first-order and moment degrees of freedom. The derivation can therefore describe the variability of the variance estimator reasonably well before its two moment approximations jointly give accurate degrees of freedom.

The distinction matters most near independence, where the population first-order MI variance is small. The quadratic sampling contribution then remains substantial relative to that population value. In Appendix A.4, $\widehat{V} \approx G/n$ and $\widehat{\nu} \approx G/2$, with G approximately following a noncentral chi-squared distribution. At independence, the approximating chi-squared distribution has mean d. The local calculation therefore predicts mean plug-in degrees of freedom near $d/2$, while the variance-estimator moment target is near d. This explains why substituting empirical probabilities into the first-order formula need not recover the sampling-moment target, even when the local approximation describes the observed component means.

Away from independence, the scaled mean errors suggest a second-order correction whose coefficient depends on the population table. A correction near d/n describes much of the mean shift in the tested 4×4 populations, but the observed variation with MI and margins requires a coefficient derived for the actual table. Such a correction would address the numerator of the moment-matching expression.

Its estimated coefficient and its effect on the completed rejection rule would still need evaluation.

7.2 Calibration and power of the MI-specific rule

Expanded Welch improves null rejection in the two selected small-sample populations. For uniform 4×4 tables at $n = 20$, it gives 0.04897, compared with 0.08352 for Wald and 0.07394 for the sample-size comparator. For 8×8 different-skew tables at $n = 100$, it gives 0.05523, compared with 0.07466 and 0.07246. Every rule is valid in all repetitions at these settings, so the reductions do not reflect failed calculations. In the wider grid, however, Expanded Welch also lowers already-conservative rates. The MI-specific cutoff improves calibration in some populations without improving it uniformly.

The paired nominal-power comparison measures the cost of those cutoffs. At two null SDs in the uniform 4×4 setting, Expanded Welch loses 2.986 percentage points relative to the sample-size comparator. After matching null rejection empirically, it instead gains 3.629 points. The two comparisons answer different questions: nominal power includes the methods' different false-positive rates, while calibrated power compares discrimination at approximately the same false-positive rate. The latter gain indicates that the sample-dependent degrees of freedom change the ordering of evidence usefully along this population path. A fixed Student t reference cannot produce that gain after calibration, because it has the same ordering by $|T|$ as Wald.

The different-skew 8×8 path also gives higher calibrated power for Expanded Welch, but its rejection curve initially falls as the true MI difference increases. At the null, the estimated difference has a positive bias; increasing $I(Q)$ makes the true signed difference negative and initially moves the estimate towards zero. Empirical null calibration changes the rejection threshold and leaves this movement in the shared statistic. The calibrated-power advantage and the initial decline are therefore both features of the tested comparison. At the larger-sample 4×4 setting, the nominal loss is small and the calibrated differences lie well within the stated practical tolerance.

7.3 Bias, uncertainty and use on observed data

The sparse 16×16 different-skew comparison demonstrates the limit of changing the cutoff. At $n = 1{,}000$, all four rules reject far above 5%. At the 0.01-nat alternative,

almost every rejection has the wrong estimated ordering. The positive difference bias is larger than the true negative difference, as shown in Equation (C.1). The four-cell construction reproduces the failure, so it is not restricted to the copying association pattern. The fixed moment reference also over-rejects: fitting denominator moments accurately does not repair a shifted numerator.

The matched same-skew populations expose a separate uncertainty problem. Their expected null difference bias cancels at equal sample sizes, but the mean squared standard error is almost twice the observed difference variance in the sparse copying setting. Rejection is strongly conservative. With unequal sample sizes, even identical population tables retain residual difference biases because each MI estimate has a different sampling distribution. The allocation results consequently depend on which population receives fewer observations, as well as on the sample-dependent cutoff.

For an observed categorical comparison, report the MI difference, its standard error and the p-value together. If Expanded Welch is used, report its estimated degrees of freedom as well. The difference supplies estimated size and direction, while the standard error and reference distribution determine the uncertainty assigned to it. A small estimated degree of freedom alone does not identify reliable calibration: it can occur near independence where Wald already rejects too rarely.

Before choosing either rejection rule for an application, evaluate it on plausible populations with the application's category definitions, sample sizes and margin patterns. Include distinct equal-MI tables and relevant alternatives, and assess false positives, power, estimated direction and validity. This comparison should guide the choice before interpreting the observed application's p-values. The online-shopping analysis illustrates the interpretation of the estimates and cutoffs, while its unknown population MI values preclude measuring calibration directly.

7.4 Limits of the evidence

The focus settings were selected using earlier calibration results. The new samples estimate their performance independently, but selection still limits the scope of the resulting power claims. Each focused curve keeps $I(P) = 0.02$ and increases $I(Q)$ along the copying family. The study does not establish the same calibrated-power advantage for reversed changes, different association locations or arbitrary equal-MI population pairs. The broad grid supplies evidence about other sample sizes, table sizes and margin regimes, rather than a universal guarantee.

The primary populations have identical row and column margins within each table. The four-cell comparison changes where dependence is concentrated while retaining MI and margins, but both constructions align the association locations between populations. Broader evidence would require unequal row and column margins, other association patterns and additional allocations. Monte Carlo intervals concern the specified populations; they do not measure performance variation over these untested populations.

The moment reference and empirical calibration require known populations. They are simulation benchmarks. Replacing Expanded Welch's sample-dependent degrees of freedom by a fixed value changes both their magnitude and their dependence on the statistic. The contrast does not isolate estimation error. Likewise, correlations and distribution-shape summaries describe the joint sampling behaviour without identifying separate causal effects. The calibrated-power bootstrap includes fitting and evaluation uncertainty for the fixed constructed alternatives, and its intervals are pointwise.

The mathematical setting assumes independent categorical observations, fixed categories, positive population cell probabilities and positive first-order MI variance. Paired samples require a covariance contribution to the difference variance. Growing alphabets and continuous MI estimators have different approximation errors. Exact independence requires the second-order argument in Appendix A.4, and the positive-MI sweep shows that proximity to that boundary also matters at finite sample sizes.

Chapter 8

Conclusion

8.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing categorical MI across independent populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, accounting for changes in cell probabilities, margins and pointwise MI. The variance of this sensitivity gives a first-order sampling-variance approximation, which is used with moment matching to choose MI-specific degrees of freedom for a Student t distribution.

The experiment evaluates the mean and sampling variance approximations separately before comparing the completed rejection rules. It contains 366 evaluation settings, with independent moment fitting at 186 null settings and 100,000 repetitions per batch. The first research question is answered by the different errors in the two moment approximations: the sampling-variance prediction can be accurate while the mean estimate remains inflated, substantially lowering the population first-order degrees of freedom relative to the moment target. Near independence, the quadratic sampling contribution explains the scale of the observed component estimates and their departure from the first-order values.

For the second research question, Expanded Welch improves false-positive rates relative to Wald and the sample-size comparator in selected small-sample populations, with a nominal power cost. At matched null rejection rates, it has higher power along the selected uniform 4×4, $n = 20$, and different-skew 8×8, $n = 100$, paths. Its calibrated differences from Wald are practically small in the different-skew 4×4, $n = 10{,}000$, comparison. These results identify a benefit of sample-dependent MI degrees of freedom beyond an overall reduction in rejection.

The method does not resolve errors in the shared statistic. In sparse different-skew tables, MI bias can shift or reverse the estimated difference and cause rejection

to fall initially as the true difference increases. Same-skew populations can instead remain conservative when the expected difference bias cancels. The matched construction and allocation checks show why population structure and sample allocation matter alongside the cutoff. Expanded Welch therefore offers useful behaviour for some categorical comparisons, while accurate estimation of the MI difference and its uncertainty remains necessary for reliable inference.

8.2 Limitations and future work

The findings concern fixed categorical populations, the tested association constructions and the specified alternative paths. They support further investigation of the MI-specific rule rather than its use as a general replacement for Normal Wald. The independently fitted references and calibration thresholds assess mechanisms under known populations; they do not supply an immediately available procedure for arbitrary observed tables.

A first extension is to derive the second-order mean contribution to the variance estimator and evaluate a corresponding degrees-of-freedom correction. A second is to improve sparse MI bias and standard-error estimation, assessing their effects separately through null rejection, power curves and estimated direction. The supporting jackknife check provides a starting point for the bias problem.

Near independence, the quadratic sampling approximation may support a more accurate comparison of MI differences. Any likelihood-ratio or resampling procedure must preserve the equal-MI null even when the complete population tables differ. Reversed power paths, additional association constructions and broader allocation comparisons would establish how far the observed calibrated-power advantage extends.

Appendix A

Supporting Calculations

These calculations support the variance sensitivity, near-independence interpretation and validity conditions in Chapter 4.

A.1 Mean of the variance influence function

The observation-level sensitivities $g_P(x, y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = P - P = 0. \tag{A.1}$$

This weighted cancellation also holds after exact plug-in substitution. Because the first derivative is linear in the direction of change, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$\begin{aligned} \mathrm{E}_P\{g_P(X, Y)\} &= \sum_{x,y} p_{xy} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\{(1-\varepsilon)P + \varepsilon\delta_{(x,y)}\}\Big|_0 && \text{(A.2)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\}\right\}\Big|_0 && \text{(A.3)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P)\Big|_0 = 0. && \text{(A.4)} \end{aligned}$$

A.2 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.5}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.6}$$

using $\log(1 + z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.7}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.8}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned} \mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i} \sum_j (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\} \\ &= \frac{t}{a_i} \sum_j H_{ij} + O(t^2) = O(t^2). \end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.29), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.29) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.9}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned} \tau^2(P_t) &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{2tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 \\ &= 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned} \tag{A.10}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.4 treats that case separately.

A.3 Dependence between the MI and variance estimates

Both $\widehat{I}(P)$ and $\widehat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\mathrm{Cov}\{\widehat{I}(P), \widehat{V}(P)\} \approx \frac{1}{n_P} \mathrm{E}_P[\{\ell_P(X,Y) - I(P)\} g_P(X,Y)]\,. \tag{A.11}$$

The corresponding expression for Q replaces P and n_P with Q and n_Q. To determine how the MI and variance estimates move together near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Appendix A.2. For small positive t, the expansions there give

$$\ell_{P_t}(i,j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \qquad g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.12}$$

Substituting both expressions into the first-order covariance above gives

$$\mathrm{Cov}\{\widehat{I}(P_t), \widehat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \tag{A.13}$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \tag{A.14}$$

The first-order covariance is therefore positive for small positive MI along this path: samples with a larger MI estimate also tend to have a larger variance estimate. In the two-sample statistic, P's and Q's MI estimates enter the numerator with opposite signs, whereas their variance contributions add under the square root in the standard error. At exact independence, $V(P_0) = 0$ and a second-order sampling approximation is needed.

A.4 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.15}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.16}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.17}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.18}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.19}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.20}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model

here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.21}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; κ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.20).

For a sampled table close to independence, the expansions in Appendix A.2, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.22}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.23}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population value $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Since $\mathrm{E}(G) = d + \kappa$ and $\mathrm{Var}(G) = 2(d + 2\kappa)$,

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.24}$$

Keeping both terms in the moment-matching equation gives

$$\nu_P^{\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.25}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.47). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.26}$$

This explains the scale of the estimated degrees of freedom and the roughly twofold plug-in overprediction in Table 6.1. The population formula alone omits the quadratic sampling contribution responsible for both effects.

A.5 Combined degrees of freedom with a zero variance estimate

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{A.27}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. The implementation nevertheless requires both components to have positive, finite degrees of freedom, so it records this case as invalid.

A.6 Numerical derivative checks

The analytic derivatives in Equations (4.18) and (4.29) were checked by centred finite differences along the one-cell perturbation paths in Chapter 4. Two positive tables,

$$\begin{pmatrix} 0.36 & 0.14 \\ 0.09 & 0.41 \end{pmatrix}, \qquad \begin{pmatrix} 0.28 & 0.09 & 0.07 \\ 0.08 & 0.17 & 0.06 \\ 0.04 & 0.10 & 0.11 \end{pmatrix},$$

were checked at every cell with step size $\varepsilon = 10^{-6}$. For each path, the difference between the values at $+\varepsilon$ and $-\varepsilon$ was divided by 2ε; all perturbed probabilities remained positive. The maximum absolute error was of order 10^{-10} for both derivatives.

Appendix B

Experimental Details

This appendix specifies the population formulas, target-MI search, complete parameter grids, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment.

B.1 Copying construction and target selection

For a positive marginal vector a, the joint distribution is

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}. \tag{B.1}$$

Summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.2}$$

Summing a column similarly gives a_j. The copying branch contributes only to its matching category, while the independent branch contributes across every category. Thus the construction preserves both margins and has positive cells whenever $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.3}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes

the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.4}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint, including the alternatives scaled by the fitted null standard deviation.

B.2 Four-cell supporting construction

For the supporting 4×4 and 16×16 comparisons, begin with the independence probabilities $a_i a_j$ and transfer an amount t within the first two rows and columns. The resulting table $R(t)$ has cell probabilities

$$r_{ij}(t) = \begin{cases} a_i a_j + t, & (i, j) = (1, 1) \text{ or } (2, 2), \\ a_i a_j - t, & (i, j) = (1, 2) \text{ or } (2, 1), \\ a_i a_j, & \text{all other cells.} \end{cases} \tag{B.5}$$

Every affected row and column gains and loses t, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \le t < t_{\max} = a_1 a_2. \tag{B.6}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

In a binary table the copying construction is exactly this transfer:

$$P(\lambda) = (1 - \lambda) \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix} \tag{B.7}$$

$$= \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda a_1 a_2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{B.8}$$

The second equality uses $a_1 + a_2 = 1$, giving $a_1 - a_1^2 = a_1 a_2$ and $a_2 - a_2^2 = a_1 a_2$. Hence $t = \lambda a_1 a_2$. This equivalence does not extend to the larger tables: copying changes cells throughout the table, whereas Equation (B.5) changes only four.

B.3 Complete parameter grids and sampling

Table B.1 lists the parameter combinations for each study and supporting check. Table sizes are $k \in \{2, 4, 8, 16\}$. Each population has identical row and column margins. Uniform margins give probability $1/k$ to every category; same skew uses $m_k(0.8)$ in both populations; different skew uses $m_k(0.7)$ in P and $m_k(0.8)$ in Q, with m_k defined by Equation (5.12). The same copying formula and root search apply to every primary-study setting.

Table B.1: Exact parameter sets; MI values are in nats.

Study	Settings
Equal MI	All four sizes and three margin regimes; $I(P) = I(Q) = 0.02$; equal sample sizes $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Population MI	4×4, uniform, same skew and different skew; $I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}$; $n \in \{100, 1000, 10000\}$.
Power	All four sizes with same skew and different skew, plus uniform 4×4; $I(P) = 0.02$, $I(Q) = 0.02 + \lvert\Delta\rvert$; $\lvert\Delta\rvert \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$; $n \in \{1000, 10000\}$.
Focused power	Settings A–C in Table 5.3; $I(P) = 0.02$, $I(Q) = 0.02 + h\sigma_0$, $h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}$. The fitted null SDs defining these populations are given below.
Four-cell check	$k \in \{4, 16\}$, same skew and different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $n \in \{1000, 10000\}$; construction in Equation (B.5).

Study	Settings
Sample allocation	Copying 4×4, same skew and different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $(n_P, n_Q) \in \{(1000, 1000), (400, 1600), (1600, 400)\}$. The total sample size is 2,000 in every comparison.

The fixed null-SD constants used to construct the focused alternatives are

$$\sigma_{0,A} = 0.154966805570126 \text{ nats},$$
$$\sigma_{0,B} = 0.065137776499116 \text{ nats},$$
$$\sigma_{0,C} = 0.003174953145465 \text{ nats}.$$

For each h in the focused grid, set the target MI of Q to $0.02 + h\sigma_0$ and solve for its copying probability. The fitting uncertainty of these constants is reported in Table C.2; the constructed populations remain fixed during evaluation.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \text{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.9}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply all four rejection rules to that pair. Repeat independently 100,000 times per distinct evaluation setting. Independently generate another 100,000 pairs for each equal-MI fitting setting. The simulations use the PCG64 pseudo-random generator with master seed 20261021 and separate streams for population, setting, fitting and evaluation samples. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. In the equal-sample studies, the two corrections cancel in their difference. In the allocation check, Equation (3.7) subtracts $9/(2n_P)$ and adds $9/(2n_Q)$ to the plug-in difference. Their combined contribution is -0.0084375 nats at $(400, 1600)$ and $+0.0084375$ at $(1600, 400)$. These terms remove the leading unequal-sample bias; the diagnostics measure the remaining bias. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule, including the zero-component case in Appendix A.5.

The sample-size comparator requires the base conditions and positive finite combined degrees of freedom in Equation (5.5). The fixed reference requires the base conditions and a positive finite fitted reference value. No component-degree requirement is added to the fixed rule.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only pairs where all four methods are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.5 Bias, variance and degrees-of-freedom diagnostics

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.10}$$

where M counts those finite-valued estimates. Centre at their sample mean, rather than the true difference, so this measures variance separately from bias. Include finite-valued MI differences even when the test output is invalid. For P and Q, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated/observed variance ratio, restrict both averages to the same repetitions with a finite difference and finite squared standard error. Retain finite

zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset; the ratio is undefined if the observed variance is zero. Report its included fraction. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Component degrees-of-freedom summaries use positive finite component values, with their included fraction reported. Combined summaries use positive finite combined values. Finite zero variance estimates remain in the moment calculations even when component degrees of freedom are undefined. Percentile ranges describe the spread of those outputs, not confidence intervals for their median. These summaries expose estimation problems but do not isolate the causal contribution of each problem to rejection.

Variability of the estimated MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.10). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order. Compare the observed value with

$$\begin{aligned} \text{population prediction} &= \frac{\tau^2(P)}{n_P}, \\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}. \end{aligned} \tag{B.11}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide each prediction by the observed variability: a ratio of one means agreement, and the ratio is undefined when observed variability is zero.

Estimated direction among rejections.

For a nonzero population difference, classify each valid two-sided rejection by the sign of $\widehat{\Delta}_{\mathrm{BC}}$. The estimate has the opposite sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta < 0$; it has the correct sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta > 0$. Report the number of rejections and the number with opposite signs for each method. The fraction with opposite signs divides the opposite-sign count by the rejection count and is undefined if there are no rejections. These counts supplement the estimate of power, which includes every valid rejection regardless of direction.

B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Wilson intervals use $z = \Phi^{-1}(0.975)$ and have centre and half-width

$$\frac{\widehat{r} + z^2/(2M)}{1 + z^2/M}, \qquad \frac{z\sqrt{\widehat{r}(1-\widehat{r})/M + z^2/(4M^2)}}{1 + z^2/M}. \tag{B.12}$$

These intervals are used for both rejection and validity rates (Wilson, 1927).

For the paired method contrast, let D_s equal the Expanded Welch valid-rejection indicator minus the comparator valid-rejection indicator on pair s. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \;\pm\; 1.96 \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (D_s - \bar{D})^2 \right\}^{1/2}. \tag{B.13}$$

Use the same common-valid subset when reporting a conditional paired contrast. Cross-construction comparisons use independent simulation streams, as do different sample allocations, so their difference uncertainty is not calculated from paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.7 Combining component moment targets

The combined moment reference follows exactly from the component moments when the two samples are independent. This calculation assumes positive component means, finite component variances and a positive variance of the summed contributions. Substitute the expected variance contributions and component moment degrees of freedom into the combination formula:

$$\frac{\{\mathrm{E}\,\widehat{V}(P)/n_P + \mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{\dfrac{\{\mathrm{E}\,\widehat{V}(P)/n_P\}^2}{2\{\mathrm{E}\,\widehat{V}(P)\}^2/\operatorname{Var}\{\widehat{V}(P)\}} + \dfrac{\{\mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{2\{\mathrm{E}\,\widehat{V}(Q)\}^2/\operatorname{Var}\{\widehat{V}(Q)\}}}. \tag{B.14}$$

For either component, the square of the mean cancels:

$$\frac{\{\mathrm{E}\,\widehat{V}(P)/n_P\}^2}{2\{\mathrm{E}\,\widehat{V}(P)\}^2/\operatorname{Var}\{\widehat{V}(P)\}} = \frac{\{\mathrm{E}\,\widehat{V}(P)\}^2}{n_P^2}\,\frac{\operatorname{Var}\{\widehat{V}(P)\}}{2\{\mathrm{E}\,\widehat{V}(P)\}^2} = \frac{\operatorname{Var}\{\widehat{V}(P)\}}{2n_P^2}. \tag{B.15}$$

Applying the same cancellation to Q reduces Equation (B.14) to

$$\frac{2\{\mathrm{E}\,\widehat{V}(P)/n_P + \mathrm{E}\,\widehat{V}(Q)/n_Q\}^2}{\mathrm{Var}\{\widehat{V}(P)\}/n_P^2 + \mathrm{Var}\{\widehat{V}(Q)\}/n_Q^2} = \frac{2\{\mathrm{E}(\widehat{\mathrm{SE}}^2)\}^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)}. \tag{B.16}$$

The final equality uses independence to remove the covariance between $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. Thus matching each component and combining at its expected contribution yields the same target as matching the sum directly. This identity concerns moments; it does not assert that the sum has a scaled chi-squared distribution. Using random sample contributions instead of their expectations produces a different, sample-dependent degrees-of-freedom rule.

B.8 Moment estimation and Monte Carlo uncertainty

For a component, let $X_s = \widehat{V}_s$ for repetition s, with population sampling mean $\mu = \mathrm{E}(X_s)$ and sampling variance $\sigma^2 = \mathrm{Var}(X_s) > 0$. From M repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.17}$$

These moments retain finite zero values. Estimate the moment degrees of freedom by $2\bar{X}^2/S_X^2$. The diagnostic decomposition uses these evaluation moments, whereas the reference cutoff uses the independent fitting moments. An undefined component degree of freedom does not remove its finite variance estimate from either calculation.

To quantify uncertainty in the moment degrees of freedom, apply the multivariate delta method to $f(m, v) = 2m^2/v$. The first-order Taylor formula for a function of two arguments is

$$f(m + \Delta m, v + \Delta v) \approx f(m, v) + \frac{\partial f}{\partial m}(m, v)\Delta m + \frac{\partial f}{\partial v}(m, v)\Delta v. \tag{B.18}$$

Here the expansion point is (μ, σ^2), the estimated point is $(\bar{X}, S_X^2)$, and the differences are $\bar{X} - \mu$ and $S_X^2 - \sigma^2$. The derivatives are

$$\frac{\partial f}{\partial m} = \frac{4m}{v}, \qquad \frac{\partial f}{\partial v} = -\frac{2m^2}{v^2}. \tag{B.19}$$

Substitution gives

$$\frac{2\bar{X}^2}{S_X^2} - \frac{2\mu^2}{\sigma^2} \approx \frac{4\mu}{\sigma^2}(\bar{X} - \mu) - \frac{2\mu^2}{(\sigma^2)^2}(S_X^2 - \sigma^2). \tag{B.20}$$

The mean and variance estimation errors come from the same repetitions, so their covariance must be retained. Their first-order contributions for one repetition are estimated by

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.21}$$

Using the fitted derivatives, the contribution for the degrees-of-freedom estimate is

$$Z_s = \frac{4\bar{X}}{S_X^2}U_s - \frac{2\bar{X}^2}{(S_X^2)^2}W_s. \tag{B.22}$$

The sample SD of Z_s, divided by $\sqrt{M}$, estimates the Monte Carlo standard error. Taking the variance of the combined contribution preserves the covariance between the estimated mean and variance. The combined reference uses the corresponding weighted contributions from both independent populations. Mean and variance ratios use the same delta-method calculation; the plug-in prediction and observed variance are kept paired when both are estimated from the same outputs.

Nominal reference-rate intervals condition on the fitted degrees of freedom. Its fitting Monte Carlo standard error is reported separately. For the calibrated comparison, refitting the fixed degrees of freedom would not change the ordering by $|T|$; the calibration bootstrap therefore resamples its empirical absolute-statistic cutoff directly.

B.9 Empirical calibration and power uncertainty

For a sample-dependent rule, assign an invalid output p-value one for calibration and order the fitting p-values. The threshold is the $(\lfloor 0.05M \rfloor + 1)$th smallest value, and rejection requires a strictly smaller evaluation p-value. Every tie at the boundary is excluded. Normal Wald and the fixed reference instead share the equivalent empirical upper-tail threshold for $|T|$, with invalid fitting values placed below all valid absolute statistics. This convention gives at most 5% rejection in the fitting batch. Independent null evaluation measures the achieved rate, including any effect of discreteness.

For focused calibrated power, resample the fitting outputs and refit these thresholds in each of 1,000 bootstrap repetitions. Apply a common resampled fitting batch across all points of a curve. At each evaluation point, resample the

joint four-method decision outcomes, preserving their pairing. Use the 2.5th and 97.5th percentiles for pointwise intervals of rates and paired rate differences. The fixed reference and Normal Wald remain identical after this calibration, so refitting their moment degrees of freedom cannot change the calibrated decisions. The population tables and effect sizes in nats stay fixed throughout the bootstrap.

Appendix C

Additional Results

This appendix contains companion approximation figures and compact supporting comparisons. The four-rule comparisons use 100,000 evaluation pairs per setting. The earlier jackknife investigation is identified separately below. Complete four-rule rates, validity denominators, component moments and paired contrasts are supplied in the experiment supplement.

C.1 Companion approximation comparisons

Figures C.1 and C.2 repeat the component decomposition for uniform and same-skew margins at common MI 0.02 nats. All ratios use the definitions in Chapter 5. Identical population tables under these nulls cancel expected difference bias, while estimation error in each component remains.

Figures C.3 and C.4 compare the Appendix A.4 predictions at smaller and larger sample sizes for the unchanged different-skew MI sweep. Shading gives the fifth–95th percentile range of the positive finite estimated component degrees of freedom. It describes sampling spread, rather than uncertainty in a mean. The companion uniform and same-skew plots are supplied electronically.

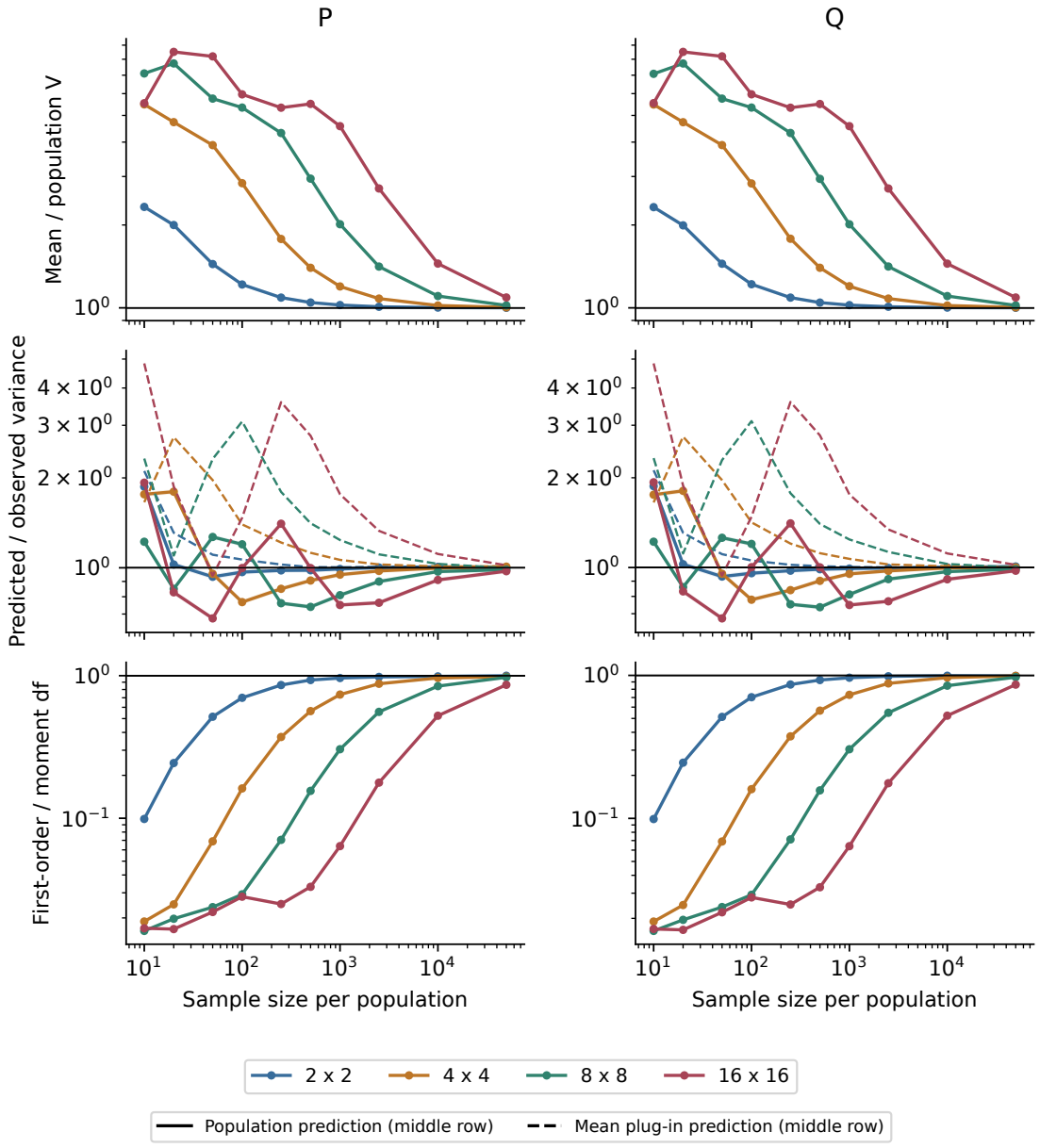


Figure C.1: Component approximation ratios under uniform margins.

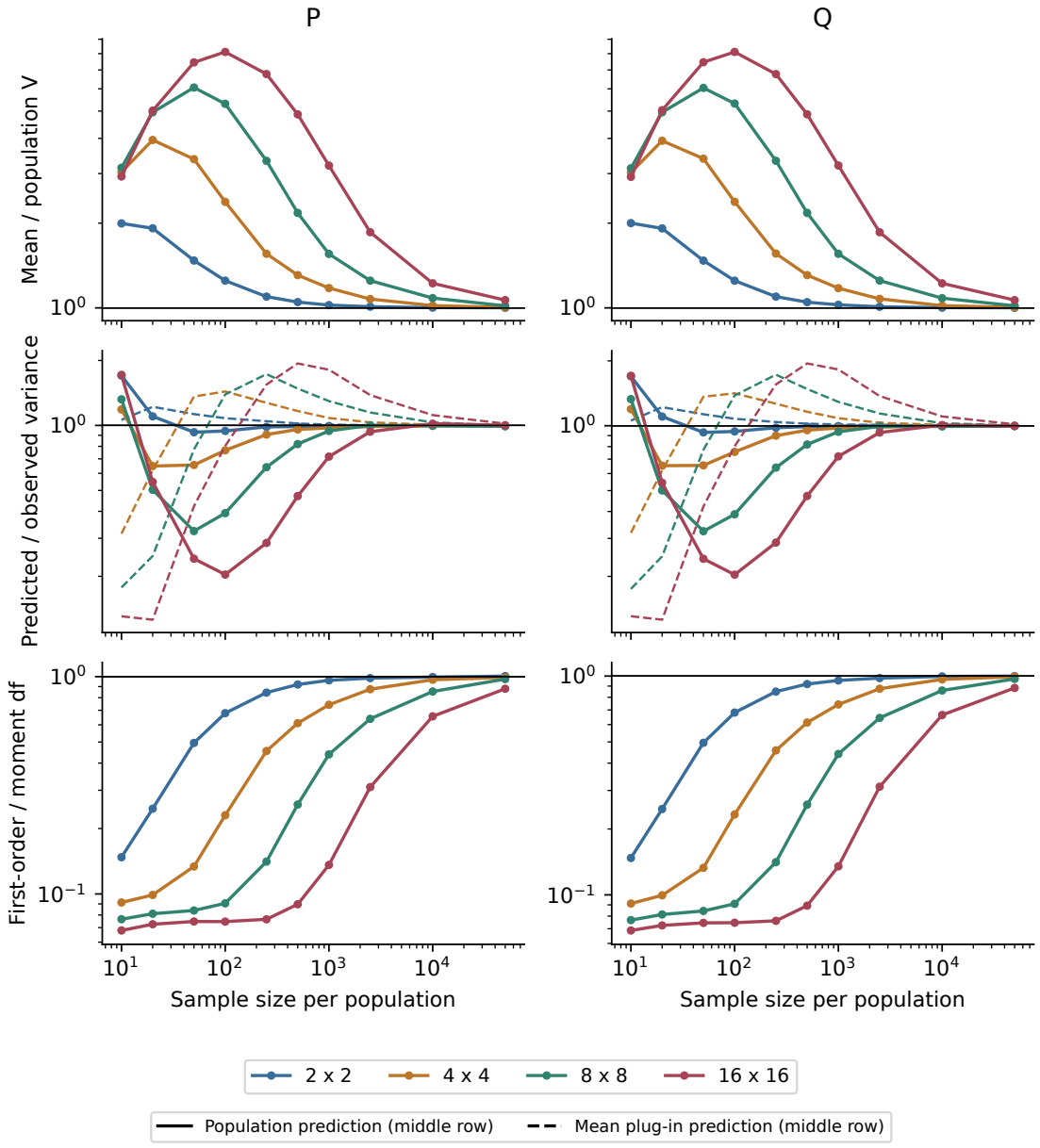


Figure C.2: Component approximation ratios under same-skew margins.

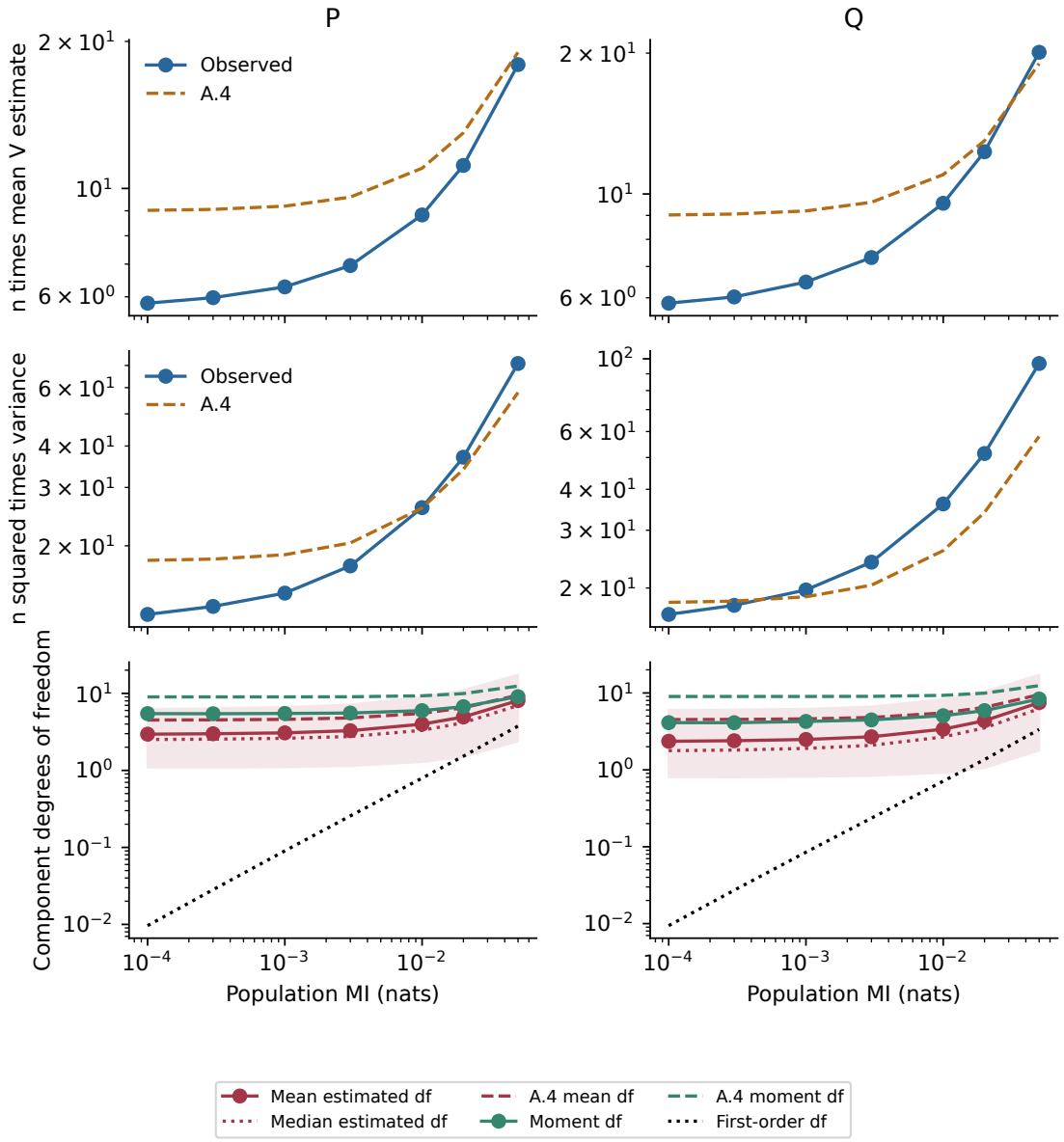


Figure C.3: Component moments and local predictions at $n = 100$.

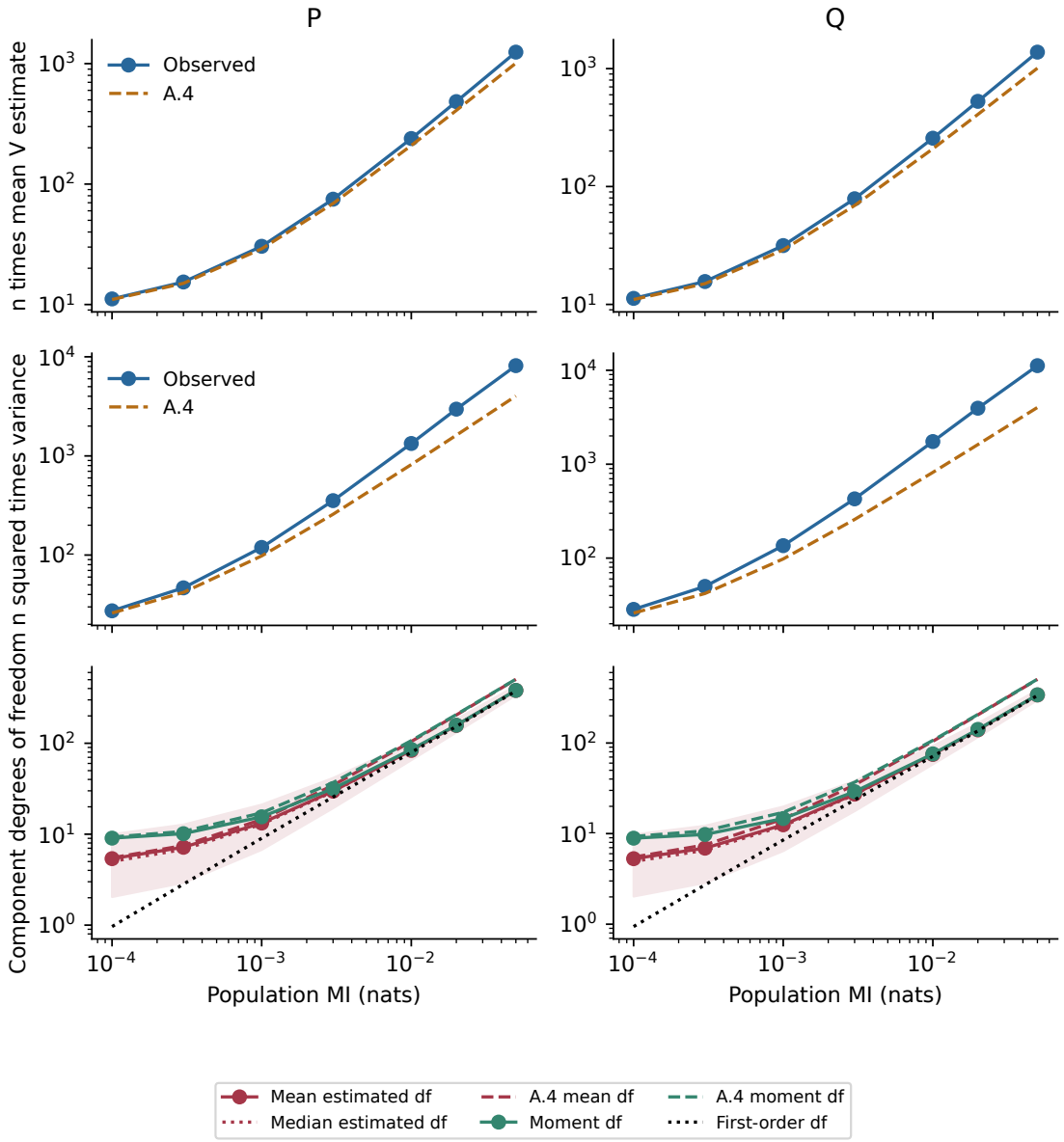


Figure C.4: Component moments and local predictions at $n = 10{,}000$.

C.2 Component estimates and reference fitting

Table C.1 reports the component means, medians and fifth–95th percentile ranges for the focus and matched failure settings. The moment target uses the evaluation sampling moments. All 100,000 component estimates are positive and finite in each shown row. Their percentile ranges describe variation across samples, not confidence intervals for a mean. The population-specific summaries complement the combined values in Chapter 6.

Table C.1: Sampling spread of component degrees of freedom.

Setting, pop.	Mean df	Median df	5th–95th	Moment df	Included
A, P	6.52	5.23	[1.95, 15.32]	14.81	100,000
A, Q	6.54	5.24	[1.93, 15.16]	14.87	100,000
B, P	14.04	12.74	[5.83, 26.60]	23.54	100,000
B, Q	9.72	8.47	[2.64, 20.91]	12.47	100,000
C, P	157.09	156.66	[131.26, 184.27]	157.96	100,000
C, Q	139.70	139.28	[115.88, 165.12]	141.27	100,000
16, same, 1,000, P	39.78	39.32	[29.18, 51.92]	71.67	100,000
16, same, 1,000, Q	39.80	39.35	[29.33, 51.88]	72.18	100,000
16, same, 10,000, P	133.18	132.82	[114.30, 153.25]	148.48	100,000
16, same, 10,000, Q	133.17	132.92	[114.19, 153.21]	146.84	100,000
16, different, 1,000, P	52.58	52.08	[41.98, 64.93]	87.28	100,000
16, different, 1,000, Q	39.80	39.33	[29.15, 51.94]	71.05	100,000
16, different, 10,000, P	172.67	172.41	[152.55, 193.72]	189.98	100,000
16, different, 10,000, Q	133.24	132.99	[114.21, 153.33]	147.07	100,000

The fixed references in Table C.2 use independent 100,000-pair fitting batches. Their Monte Carlo standard errors follow Appendix B.8. The same batches estimate the null SD used to construct each power path. The constructed differences are then held fixed, so power uncertainty is conditional on those population tables. Across all 186 fitting settings, the directly observed variance of the summed variance contributions differs from the sum of component variances by at most 0.88%, consistent with the small empirical covariances between independently sampled components.

Table C.2: Independent moment references and fitted null standard deviations.

Setting	Reference df	MCSE	Null SD (nats)	MCSE (nats)
A	29.70	0.10	0.154967	0.000363
B	33.34	0.11	0.065138	0.000152
C	293.98	0.94	0.003175	0.000007

C.3 Same-skew null comparison

Figure C.5 completes the broad null comparison. Both populations are identical under the copying construction, so their expected MI estimation biases cancel at equal sample sizes. Small-sample conservatism therefore occurs even without an expected difference bias. The lower panels show validity for Normal Wald and Expanded Welch; the other two rules share Wald's validity in these settings. Shading around the rejection curves gives pointwise 95% Wilson intervals.

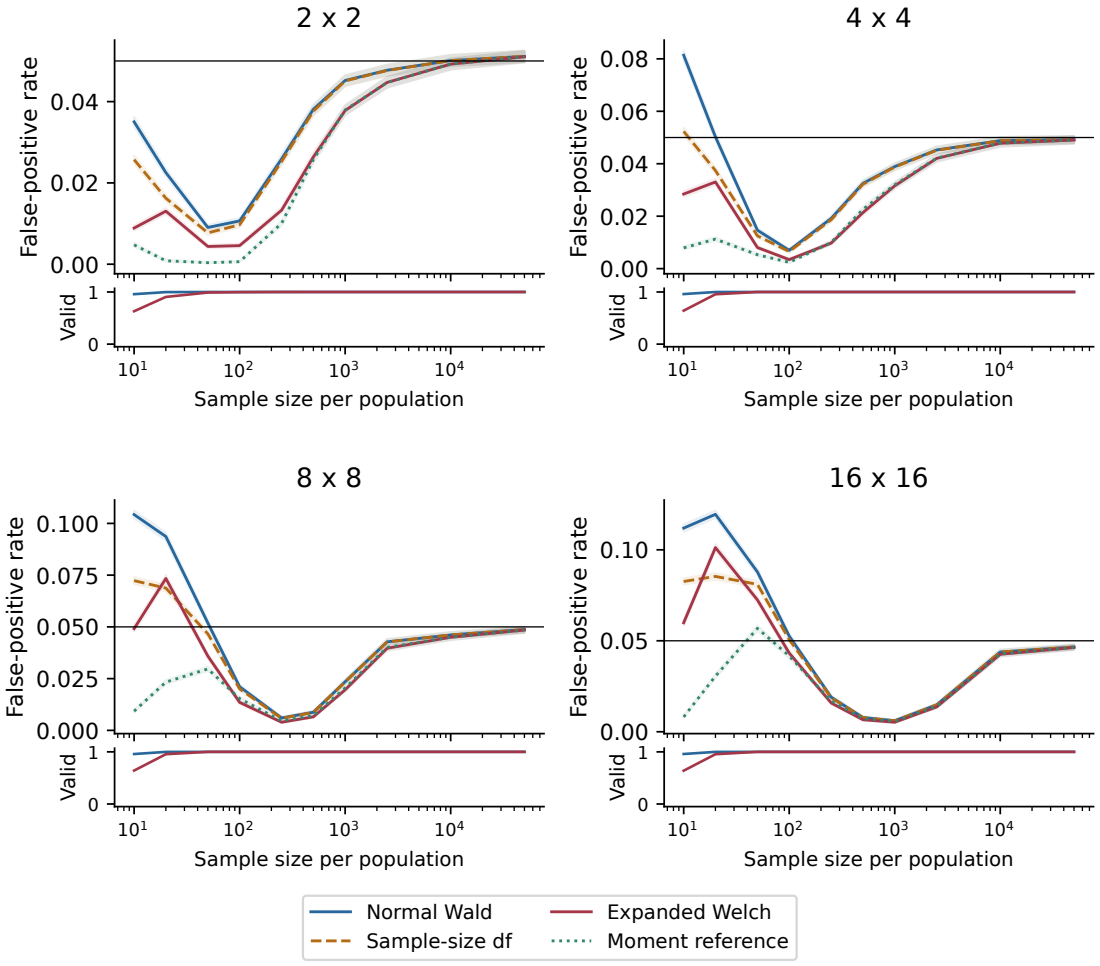


Figure C.5: False-positive rates under same-skew margins.

C.4 Broader power and estimated direction

Figures C.6 and C.7 retain the broader differences in nats at $n = 1{,}000$ and 10,000. Each curve keeps $I(P) = 0.02$ and increases $I(Q)$. They show Normal Wald and Expanded Welch; the complete four-rule results are in the supplement. Different-skew large tables show the initial decline associated with the biased estimated difference. The matched same-skew comparison does not show the same large wrong-direction rejection problem.

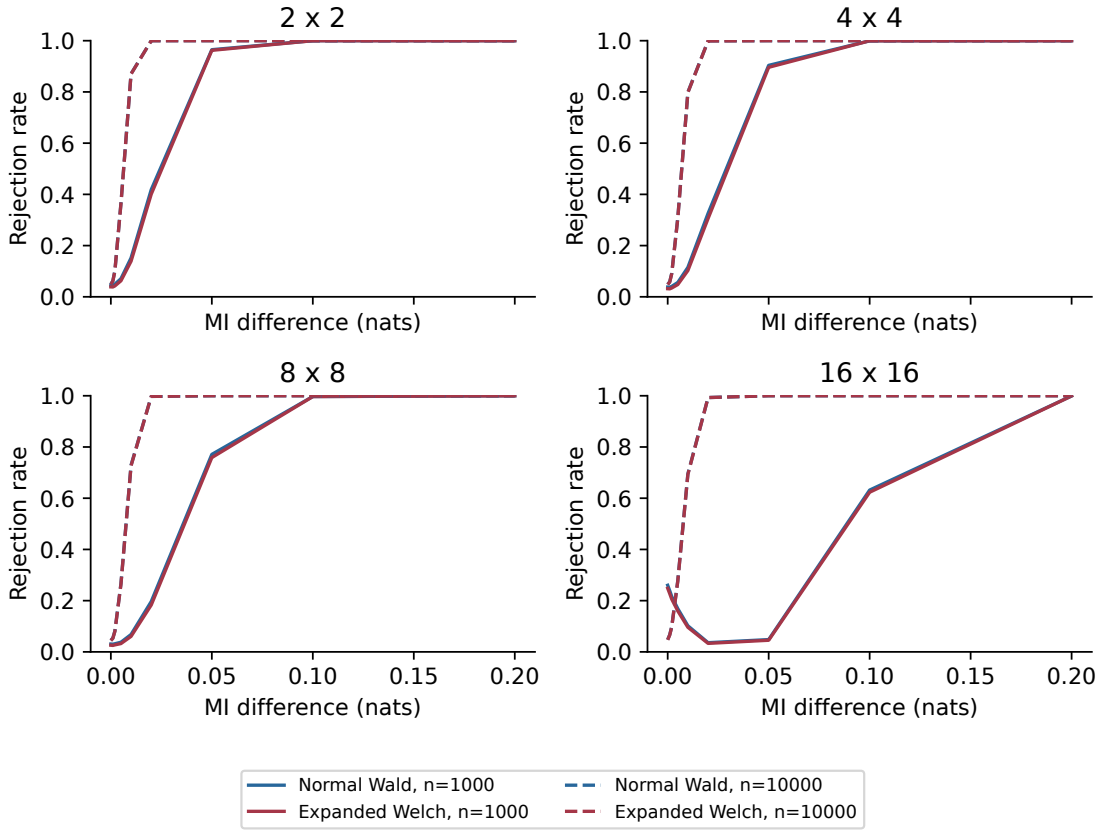


Figure C.6: Broader power comparison under different-skew margins.

At 16×16, different skew, $n = 1{,}000$ and a true 0.01-nat difference, 10,183 of 10,184 Wald rejections and 9,591 of 9,592 Expanded Welch rejections have the wrong estimated ordering. The fixed reference gives 9,829 wrong-direction cases among 9,830 rejections. Under same skew, the corresponding Wald and Expanded Welch counts are 59 of 1,809 and 52 of 1,646. At $n = 10{,}000$, none of the four rules has a wrong-direction rejection at this alternative under either skew regime. The different-skew alternative has true difference -0.01 nats, but its difference bias is 0.03071 nats at $n = 1{,}000$. Thus

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx -0.01 + 0.03071 = 0.02071 \text{ nats.} \tag{C.1}$$

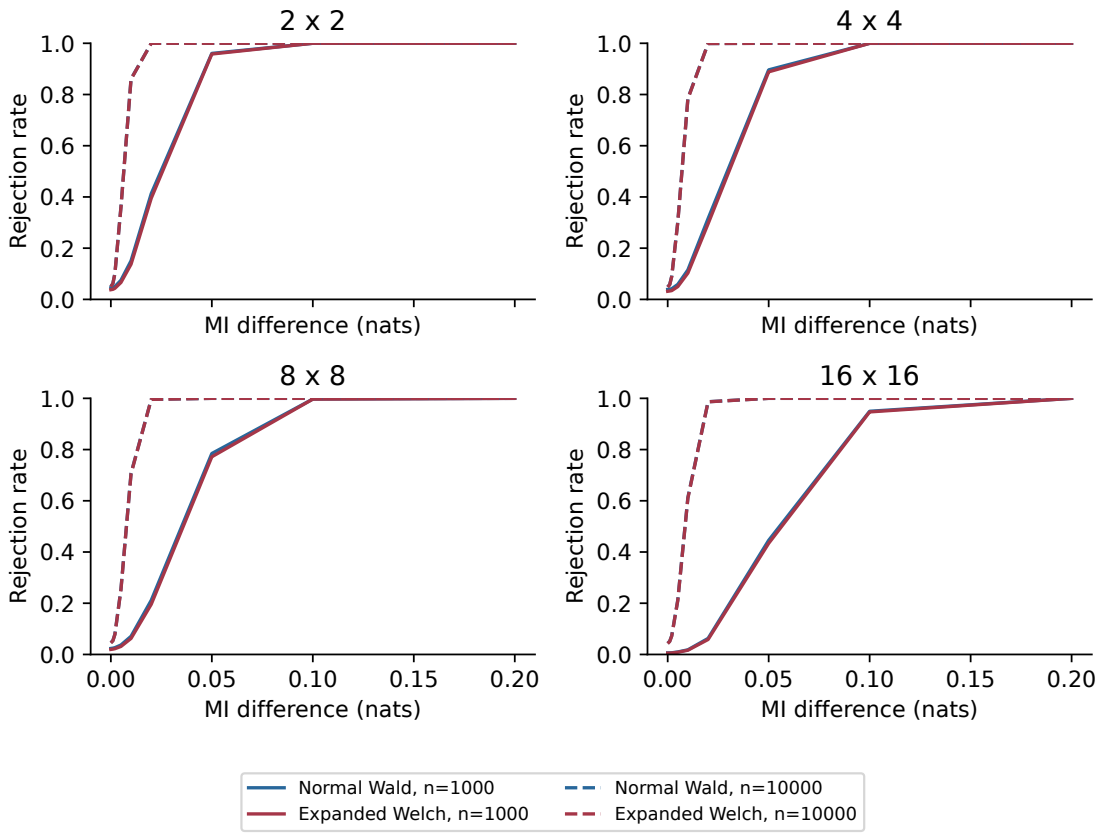


Figure C.7: Matched broader power comparison under same-skew margins.

The positive centre explains why almost every rejection indicates the wrong ordering. Two-sided power retains every rejection; these counts show why its interpretation also requires the estimated direction.

C.5 Matched construction comparison

Table C.3 compares null rejection under the copying and four-cell constructions. Each row uses common MI 0.02 nats, the margin vectors in Chapter 5, and equal sample sizes. The first column identifies table size, skew regime, construction and sample size. The four-cell construction follows Equation (B.5). Alternative rates and the corresponding diagnostics are retained in the supplement.

Table C.3: Matched null rejection under copying and four-cell constructions.

Setting	Wald	Size df	Expanded	Reference
4, Different, copy, 1000	0.0381	0.0379	0.0316	0.0320
4, Different, copy, 10000	0.0486	0.0486	0.0477	0.0477
4, Different, four, 1000	0.0387	0.0385	0.0321	0.0322
4, Different, four, 10000	0.0488	0.0487	0.0480	0.0480
4, Same, copy, 1000	0.0389	0.0387	0.0316	0.0320
4, Same, copy, 10000	0.0488	0.0488	0.0478	0.0479
4, Same, four, 1000	0.0393	0.0392	0.0325	0.0325
4, Same, four, 10000	0.0491	0.0491	0.0481	0.0480
16, Different, copy, 1000	0.2601	0.2595	0.2484	0.2535
16, Different, copy, 10000	0.0478	0.0478	0.0468	0.0469
16, Different, four, 1000	0.3096	0.3090	0.2972	0.3013
16, Different, four, 10000	0.0511	0.0511	0.0501	0.0501
16, Same, copy, 1000	0.0060	0.0060	0.0053	0.0056
16, Same, copy, 10000	0.0438	0.0438	0.0426	0.0427
16, Same, four, 1000	0.0113	0.0112	0.0100	0.0104
16, Same, four, 10000	0.0472	0.0471	0.0461	0.0462

C.6 Unequal allocation at a fixed total

Table C.4 fixes the 4×4 population pairs and changes the allocation of 2,000 observations. The first column gives skew regime, n_P/n_Q and the absolute MI difference in nats. Null rows use $I(P) = I(Q) = 0.02$, while alternative rows use $I(P) = 0.02$ and $I(Q) = 0.03$. Each allocation has its own independently fitted null reference. Reversing the allocation keeps the population labels and tables fixed.

The allocations change both estimation bias and uncertainty. Even the same-skew null can retain unequal residual biases when its sample sizes differ. These comparisons assess the full procedure under imbalance. The moment identity for combining independent components remains exact; its accuracy is distinct from the

Table C.4: Nominal rejection under balanced and unequal allocations.

Setting	Wald	Size df	Expanded	Reference
Different, 400/1600, 0.00	0.0450	0.0446	0.0293	0.0325
Different, 1000/1000, 0.00	0.0381	0.0379	0.0316	0.0320
Different, 1600/400, 0.00	0.0452	0.0449	0.0251	0.0297
Different, 400/1600, 0.01	0.1324	0.1319	0.1037	0.1070
Different, 1000/1000, 0.01	0.1148	0.1146	0.1026	0.1018
Different, 1600/400, 0.01	0.0532	0.0523	0.0342	0.0290
Same, 400/1600, 0.00	0.0465	0.0461	0.0270	0.0311
Same, 1000/1000, 0.00	0.0389	0.0387	0.0316	0.0320
Same, 1600/400, 0.00	0.0462	0.0458	0.0263	0.0310
Same, 400/1600, 0.01	0.1277	0.1273	0.0917	0.0971
Same, 1000/1000, 0.01	0.1150	0.1147	0.1028	0.1012
Same, 1600/400, 0.01	0.0520	0.0512	0.0349	0.0301

sampling behaviour of the estimated weights and the distributional approximation for the statistic.

C.7 A jackknife check of MI difference bias

This earlier supporting check uses 20,000 pairs per setting, separately from the 100,000-pair experiment. It asks whether changing the bias correction improves the sparse MI differences that give wrong-direction rejections in the main results. The delete-one jackknife uses the change in an estimate after removing one observation to estimate its leading finite-sample bias (Paninski, 2003, Section 3).

For the sample from P, let $\widehat{I}^{(-a)}(P)$ be the raw plug-in MI after removing observation a. Each such estimate uses the remaining $n_P - 1$ observations and recalculates both margins. Subtracting $n_P - 1$ times the change between the average leave-one-out estimate and the full-sample estimate gives

$$\begin{aligned}\widehat{I}_{\mathrm{J}}(P) &= \widehat{I}(P) - (n_P - 1)\left\{\frac{1}{n_P}\sum_{a=1}^{n_P}\widehat{I}^{(-a)}(P) - \widehat{I}(P)\right\} \\ &= n_P\widehat{I}(P) - \frac{n_P - 1}{n_P}\sum_{a=1}^{n_P}\widehat{I}^{(-a)}(P). \end{aligned} \tag{C.2}$$

Because observations in the same cell give identical leave-one-out tables, the average can be calculated once per occupied cell, weighting its leave-one-out MI by N_{ij}^P/n_P. A cell reduced to zero contributes zero to the MI sum. Apply the same calculation to Q, retaining all row and column categories and any negative corrected estimates.

The jackknife replaces the analytic correction in Equation (3.7). The estimated difference and statistic become

$$\widehat{\Delta}_{\mathrm{J}} = \widehat{I}_{\mathrm{J}}(P) - \widehat{I}_{\mathrm{J}}(Q), \qquad T_{\mathrm{J}} = \frac{\widehat{\Delta}_{\mathrm{J}}}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}}. \tag{C.3}$$

The variance estimates in the denominator are unchanged. Compare $|T_{\mathrm{J}}|$ with the 97.5th percentile of the standard normal distribution, approximately 1.96, giving the same two-sided 5% cutoff as Wald. This separates changing the MI estimate from changing its estimated uncertainty.

The four settings use 16×16 copying populations with margins $m_{16}(0.7)$ for P and $m_{16}(0.8)$ for Q, as defined in Equation (5.12). They fix $I(P) = 0.02$ and use $I(Q) = 0.02$ or 0.03 nats at $n_P = n_Q = 1{,}000$ and 10,000. Both corrections are applied to the same table pairs in each setting.

Table C.5: Analytic and jackknife MI differences with the same standard error.

n	Δ	Correction	Bias	Obs. SD	Est./obs.	Rejection rate
1,000	0.00	Analytic	+0.02960	0.01449	1.94	0.2522
1,000	0.00	Jackknife	-0.00116	0.01521	1.76	0.0085
1,000	-0.01	Analytic	+0.03061	0.01564	1.81	0.1005
1,000	-0.01	Jackknife	-0.00024	0.01638	1.65	0.0275
10,000	0.00	Analytic	-0.00088	0.00387	1.07	0.0463
10,000	0.00	Jackknife	-0.00045	0.00401	1.00	0.0507
10,000	-0.01	Analytic	-0.00094	0.00433	1.05	0.6992
10,000	-0.01	Jackknife	-0.00057	0.00447	0.99	0.6619

Table C.5 shows a much smaller difference bias at $n = 1{,}000$. Under equal MI, it falls from 0.02960 to -0.00116 nats; under the alternative, from 0.03061 to -0.00024 nats. The individual jackknife estimates remain upward biased. Their null biases are 0.03439 nats for P and 0.03555 for Q. Under the alternative, the jackknife MI biases are 0.03452 for P and 0.03476 nats for Q. These similar biases largely cancel in the difference. The observed standard deviation increases slightly, but the root mean squared error falls from 0.03295 to 0.01525 nats under the null and from 0.03438 to 0.01638 under the alternative.

The jackknife difference also gives a more accurate estimated ordering at the sparse alternative. Opposite-direction rejections fall from 2,009 to 5, while correct-direction rejections rise from 1 to 544. The total rejection rate falls because the original rejections almost all indicate the wrong ordering. Null rejection, however, falls to 0.0085, with a Wilson 95% Monte Carlo interval of [0.0073, 0.0098]. The mean squared standard error still exceeds the observed variance of the jackknife difference: their ratio is 1.76 under the null and 1.65 under the alternative. Better estimation of the difference therefore leaves a separate problem in estimating its uncertainty.

At $n = 10{,}000$, the jackknife null rejection rate is 0.0507, with a Wilson interval of [0.0477, 0.0538]. The estimated/observed variance ratios for the analytic and jackknife differences are both close to one. Alternative rejection falls from 0.6992 to 0.6619, with every rejection indicating the correct direction for both corrections. The jackknife difference has less bias but slightly greater observed variability and root mean squared error at this sample size.

C.8 Computational cost

Table C.6 compares scalar-call times, measured separately from sampling and population construction. It uses 200 pre-generated different-skew null pairs at $n = 1{,}000$ for each table size. After 20 warm-up calls per method, five timed passes through those pairs give 1,000 calls per method and setting. The call order alternates between passes. Measurements were made on an arm64 machine with macOS 26.6.2, Python 3.13.5, NumPy 2.4.4 and SciPy 1.17.1. These measurements concern Normal Wald and Expanded Welch; they do not time the two additional simulation benchmarks.

Table C.6: Time per scalar test call in microseconds: median [interquartile range].

Size	Normal Wald	Expanded Welch
2×2	101.83 [99.25, 104.87]	186.58 [182.46, 191.67]
4×4	107.85 [101.92, 113.51]	197.13 [188.83, 205.25]
8×8	103.46 [101.12, 107.30]	191.81 [187.00, 197.26]
16×16	112.42 [108.33, 116.43]	202.29 [198.07, 208.00]

Expanded Welch takes approximately 1.8–1.9 times as long in these measurements. Both methods require calculations over the table cells, and the fixed call overhead remains material at these dimensions.

C.9 Holm adjustment for the application

Table C.7 gives the raw and adjusted p-values for all six comparisons used to select the online-shopping application. Each tests equal predictor–outcome MI between two groups. Adult compares female and male records using education (16 categories) or occupation (14 categories) and income above US$50,000 (Becker and Kohavi, 1996). Its training and test partitions are combined without survey weights; records with missing occupation are excluded only from that comparison. Student Performance compares schools GP and MS using Portuguese-language course records, study time (four levels) or maternal education (five levels), and a final grade of at least 10 (Cortez, 2008). The shopping comparisons use month or weekend and purchasing among new and returning visitors.

Table C.7: Raw and Holm-adjusted p-values for the six exploratory comparisons.

	Normal Wald		Expanded Welch	
Comparison	Raw	Adjusted	Raw	Adjusted
Adult: occupation	1.18×10^{-44}	7.10×10^{-44}	1.64×10^{-42}	9.81×10^{-42}
Adult: education	1.89×10^{-34}	9.45×10^{-34}	4.93×10^{-33}	2.46×10^{-32}
Online Shoppers: month	4.71×10^{-5}	1.89×10^{-4}	3.73×10^{-4}	1.49×10^{-3}
Student Performance: study time	0.1982	0.5946	0.2339	0.7016
Student Performance: maternal education	0.8361	1	0.8468	1
Online Shoppers: weekend	0.9554	1	0.9601	1

Holm's multiple-testing adjustment is applied separately to each method's six p-values (Holm, 1979). Order the raw p-values from smallest to largest and multiply them by 6, 5, 4, 3, 2 and 1, respectively. Moving through that order, replace any value smaller than the preceding adjusted value by that preceding value, and cap the results at one. The month–purchase comparison has the third-smallest raw p-value for both methods, and the earlier adjusted values are smaller, so its adjusted p-value is four times its raw value. Both adjusted month p-values remain below 0.05.

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